

TERM 1 / CELLS AND MOLECULES

Biochemistry Retake

Review • Practice • Source papers

Original confirmed Term 1 syllabus: Lippincott Chapters 1-7, 14-18 and 23-33, plus foundations, water/buffers and laboratory-derived theory. Retake-specific exclusions have not been announced here. Chapters 8-13 and 19-22 are not standalone sections. Physiology, histology and the Term 2 metabolism course are excluded.

How to use this review

Start with the chapter explanation and comparison table. Answer a short practice set, then use its review mapping to repair weak points. Guided mode opens the matching paragraph after you answer; unguided mode leaves the reference closed. Past-paper checkpoints explain the tested relationship, including defects in the original wording.

What is and is not included

The scope follows the latest confirmed original Term 1 record. A narrower retake announcement may change it. Integrated metabolism is retained where explicitly confirmed in Chapters 14-18 and 23-27; this is not the separate Biochemistry II pathway bank. Laboratory-derived theory is included as a dedicated section.

Past-paper integrity

Four biochemistry sections contain 242 original question occurrences. The alternate February copy is treated as a duplicate. Some displayed choices are normalized; study answer letters need not match the source PDF order. Invalid source items are explicitly repaired or accept more than one scientifically valid answer. Study keys are not certified official keys.

Edition and provenance

Lippincott Illustrated Reviews: Biochemistry, Ferrier, sixth edition, plus the original teacher slides and laboratory notes. Each concept cites its source. This is educational review, not patient-specific clinical guidance. Version: 2026-09-30

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01 Biochemical foundations

Source basis: Teacher foundations / laboratory material; concept-specific teacher references below.

What biochemistry explains in medicine

Biochemistry applies chemical principles to biological processes at the molecular and cellular levels. It connects a molecular change to altered cell function, a clinical phenotype, a diagnostic marker, and sometimes a targeted treatment. A useful medical answer therefore moves from molecule to mechanism to patient rather than stopping at a definition.

- Biochemistry studies the chemistry of living systems.
- Molecular abnormalities can explain disease mechanisms and laboratory findings.
- Mechanistic understanding supports diagnosis, prevention, and rational treatment.

Application: A pathogenic enzyme variant can change a metabolite concentration and create a recognizable disease pattern. Biomarkers are clinically useful because they report an underlying molecular or cellular process.

Sources: INTRODUCTION.ppt, slides 3-6 (definition, scope, and medical relevance). biochemistry-scope.md, Primary course files: INTRODUCTION.ppt.

Elements and major biomolecule classes

Most biological matter is built chiefly from carbon, hydrogen, nitrogen, oxygen, phosphorus, and sulfur. Carbohydrates, lipids, proteins, and nucleic acids have distinct recurring chemical features and major roles, but they interact in cells rather than operating as isolated lists. Proteins perform much of the cell's catalytic and structural work, whereas nucleic acids store and transmit hereditary information.

- CHNOPS denotes carbon, hydrogen, nitrogen, oxygen, phosphorus, and sulfur.
- Carbohydrates commonly provide fuel and participate in structure and recognition.
- Lipids store energy, form membranes, insulate, and generate signaling molecules.
- Proteins act as enzymes, receptors, transporters, antibodies, and structural elements.
- Nucleic acids store and express genetic information.

Application: Different biomolecule classes require different laboratory measurements and therapeutic approaches.

Sources: INTRODUCTION.ppt, slides 9-13 (CHNOPS, cells, biomolecule classes, and central dogma).

Central dogma and genotype-to-phenotype reasoning

The usual flow of sequence information is DNA to RNA to protein. A DNA variant can alter an RNA sequence or its expression and thereby change the amount or amino-acid sequence of a protein. The resulting change in protein structure or function is the mechanistic bridge from genotype to phenotype.

- Transcription transfers information from DNA to RNA.
- Translation uses messenger RNA to specify a polypeptide sequence.
- Mutations can affect protein sequence, abundance, localization, or regulation.

Application: Missense variants can replace one amino acid and change enzyme or hemoglobin function. Some disease variants affect RNA processing or expression without changing a coding sequence.

Sources: INTRODUCTION.ppt, slide 13 (central dogma and information flow).

Anabolism, catabolism, and ATP coupling

Catabolic pathways break molecules down and often capture released energy, whereas anabolic pathways build molecules and require energy and reducing power. ATP couples favorable energy-releasing reactions to otherwise unfavorable cellular work through group transfer and linked reactions. ATP is an immediate energy currency, not the body's only energy store.

- Catabolism is degradative and commonly energy yielding.
- Anabolism is synthetic and commonly energy requiring.
- ATP contains adenine, ribose, and three phosphate groups.
- ATP hydrolysis can be coupled to biosynthesis, transport, and mechanical work.

Application: Hypoxia impairs ATP generation and rapidly compromises ion pumps and cell viability.

Sources: INTRODUCTION.ppt, slides 16-19 (anabolism, catabolism, and ATP).

Functional groups determine biochemical behavior

Functional groups give biomolecules predictable acid-base behavior, polarity, reactivity, and interaction patterns. Hydroxyl groups are polar; carboxyl and amino groups participate in proton transfer; carbonyl groups define aldehydes and ketones; phosphate groups carry charge and transfer energy; and sulfhydryls can form disulfides. Recognizing the group is more useful than memorizing an isolated molecule because the same chemistry reappears in proteins, carbohydrates, lipids, and nucleotides.

- Hydroxyl groups support polarity and hydrogen bonding.
- Carboxyl groups can donate protons; amino groups can accept protons.
- Carbonyl groups occur in aldehydes and ketones.
- Phosphate groups contribute negative charge and participate in energy and signal transfer.
- Two oxidized cysteine sulfhydryls can form a disulfide bond.

Application: Ionization of functional groups affects drug solubility, membrane passage, and protein binding.

Sources: INTRODUCTION.ppt, slides 20-22 (major biological functional groups).

Covalent and noncovalent forces in biomolecules

Covalent bonds share electrons and build stable molecular backbones, while ionic interactions, hydrogen bonds, and hydrophobic effects organize molecules and are often reversible. Unequal electron sharing makes a bond polar and creates partial charges that can interact with water. Nonpolar groups cluster in water because this minimizes disruption of water-water hydrogen bonding, a major driver of membranes and protein folding.

- Ionic interactions are attractions between opposite charges.
- Polar covalent bonds share electrons unequally; nonpolar covalent bonds share them more evenly.
- Hydrogen bonds are individually weak but collectively important.
- Disulfide bonds are covalent links between cysteine residues.
- The hydrophobic effect drives nonpolar groups away from water.

Application: Heat, extreme pH, and detergents can disrupt noncovalent forces and denature proteins. A mutation that replaces a charged residue with a hydrophobic one can promote abnormal protein aggregation.

Sources: INTRODUCTION.ppt, slides 24-31 (ionic/covalent bonds, polarity, disulfides, hydrogen bonds, and hydrophobic interactions).

Past-paper checkpoints and wording traps

Which substance is produced in every dehydration synthesis reaction?

Water is the best answer. Condensation joins monomers while releasing water.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q30

Formic acid is also called

Methanoic acid is the best answer. The IUPAC name for one-carbon formic acid is methanoic acid.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q60

The general formula of amines is

R-NH₂ is the best answer. A primary amine has the general form R-NH₂.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q1

If you replace the oxygen of the hydroxyl group with sulfur in R-OH, which functional group will you get?

Thiol is the best answer. Replacing O in an alcohol with S gives a thiol, R-SH.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q14

02 Water, acids, bases and buffers

Source basis: Teacher foundations / laboratory material; concept-specific teacher references below.

Calculation toolkit

Relationship	Use / trap
$\text{pH} = -\log_{10}[\text{H}^+]$	One pH unit is a tenfold change in hydrogen-ion activity (concentration approximation in dilute solutions).
$\text{pH} = \text{pK}_a + \log_{10}([\text{A}^-]/[\text{HA}])$	At equal conjugate base and weak acid, $\text{pH} = \text{pK}_a$. Use concentrations after any strong acid/base neutralization.
Buffering range $\text{pK}_a \pm 1$	A useful rule of thumb, not an abrupt cutoff. Higher total concentration increases buffer capacity at the same ratio.
0.070 M lactate / 0.080 M lactic acid; pK_a 3.74	$3.74 + \log_{10}(0.875) = 3.68$. Concentration ratio is base/acid, not acid/base.
pI for an amino acid	Average the two pK_a values that bracket the electrically neutral species. Do not blindly average all pK_a values.

Body-water compartments and ionic composition

Total body water is distributed mainly between intracellular fluid and extracellular fluid, with roughly two thirds intracellular. Extracellular fluid includes plasma and interstitial fluid, while transcellular fluids are specialized small compartments such as cerebrospinal fluid. Sodium and chloride predominate extracellularly, whereas potassium, magnesium, and phosphate are relatively enriched intracellularly.

- About two thirds of total body water is intracellular.
- ECF comprises plasma, interstitial fluid, and small transcellular compartments.
- Sodium is the major ECF cation; potassium is the major ICF cation.
- Solute distribution and membrane permeability govern water movement.

Application: Fluid therapy changes ECF first and then redistributes according to tonicity and permeability. Serum sodium is primarily a marker of water balance relative to exchangeable body solute.

Sources: Water and Buffer 1404.ppt, slides 4-5 (body water and compartments). Marks water/buffer reference, local reference pages 2-4 (compartments, electrolytes, and osmolality).

Water polarity, hydrogen bonding, and solubility

Water is polar because oxygen attracts shared electrons more strongly than hydrogen and the molecule has a bent geometry. Its partial charges permit hydrogen bonding and hydration shells around ions and polar solutes. Amphipathic molecules contain both hydrophilic and hydrophobic regions and therefore organize into structures such as membranes and micelles.

- Water is a dipole with partial negative charge on oxygen and partial positive charge on hydrogen.
- Hydrogen bonding explains much of water's solvent behavior.
- Hydrophilic solutes interact favorably with water; hydrophobic solutes do not.
- Amphipathic molecules contain both polar and nonpolar regions.

Application: Amphipathic bile salts emulsify dietary lipids. Drug polarity influences distribution between aqueous compartments and membranes.

Sources: Water and Buffer 1404.ppt, slides 4 and 6-8 (water polarity, hydrogen bonds, and solubility classes).

pH, pOH, and logarithmic concentration reasoning

pH is the negative base-10 logarithm of hydrogen-ion concentration, so each pH unit represents a tenfold change in hydrogen-ion concentration. At 25°C, pH plus pOH equals 14, which allows conversion between hydrogen and hydroxide concentrations. Because physiology operates over a narrow pH range, small numerical pH shifts represent substantial chemical changes.

- $\text{pH} = -\log_{10}[\text{H}^+]$.
- At 25°C, $\text{pH} + \text{pOH} = 14$.
- A fall of one pH unit means a tenfold increase in $[\text{H}^+]$.
- Acidemia and alkalemia refer to measured blood pH, not the underlying process.

Application: Arterial pH below the reference range is acidemia; above it is alkalemia. A change from pH 7.4 to 7.1 is chemically much larger than the decimal difference suggests.

Sources: Water and Buffer 1404.ppt, slides 9-15 (pH, pOH, clinical pH, acidosis, and alkalosis).

Acids, bases, conjugate pairs, Ka, and pKa

A Brønsted acid donates a proton and a Brønsted base accepts one; loss of a proton converts an acid into its conjugate base. Strong acids dissociate more completely than weak acids in water. A larger Ka and therefore a lower pKa indicate a stronger acid and make quantitative prediction of ionization possible.

- Acid and conjugate base differ by one proton.
- Ka measures the extent of acid dissociation.
- $\text{pKa} = -\log_{10} \text{Ka}$.
- Lower pKa means stronger acid.
- Weak does not mean dilute; it describes incomplete dissociation.

Application: The pKa of a drug helps predict its charge state and membrane permeability at a given pH.

Sources: Water and Buffer 1404.ppt, slides 14 and 17-20 (acid-base definitions, conjugate pairs, Ka, and pKa).

Henderson-Hasselbalch equation and ionization ratios

The Henderson-Hasselbalch equation is $\text{pH} = \text{pKa} + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$. It converts the difference between pH and pKa into the ratio of conjugate base to weak acid: one unit above pKa means 10:1, and one unit below means 1:10. At pH equal to pKa, acid and conjugate base are equal.

- $\text{pH} - \text{pKa} = \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$.
- $\text{pH} = \text{pKa}$ gives $[\text{A}^-] = [\text{HA}]$.
- Each unit of pH-pKa changes the ratio tenfold.
- Above pKa the deprotonated form predominates; below pKa the protonated form predominates.

Application: Ion trapping of weak acids and bases follows the same pH-pKa logic. The bicarbonate equation applies this relationship to blood acid-base physiology.

Sources: Water and Buffer 1404.ppt, slides 24-28 and teacher MCQs 37-38. Lippincott 6e, Chapter 1 acid-base section, local PDF pages 18-21.

Titration curves and how buffers resist pH change

A buffer contains a weak acid and its conjugate base, allowing added H^+ to be consumed by base and added OH^- to be consumed by acid. Its flattest titration-curve region lies near pKa, with greatest effectiveness roughly within pKa plus or minus one pH unit. Buffer capacity depends on the total concentration of the pair, not just their ratio.

- At half-equivalence, $\text{pH} = \text{pKa}$ and buffering is maximal.
- The useful buffer range is approximately $\text{pKa} \pm 1$.

- The conjugate base consumes added H^+ .
- Higher total buffer concentration gives greater capacity at the same pH and ratio.

Application: A patient's ability to buffer an acid load depends on available buffer stores as well as ventilation and renal response.

Sources: Water and Buffer 1404.ppt, slides 22-28 (titration, buffer region, range, and capacity).

Bicarbonate, phosphate, proteins, and hemoglobin in acid-base control

The major physiological buffers are bicarbonate, phosphate, plasma proteins, and hemoglobin. Bicarbonate is an open system because the lungs regulate CO_2 rapidly and the kidneys regulate bicarbonate and net acid more slowly. Protein side chains, especially in hemoglobin, accept or donate protons; compensation limits a pH disturbance but does not remove its primary cause.

- Bicarbonate couples blood chemistry to pulmonary CO_2 elimination.
- Phosphate is important intracellularly and in urine.
- Ionizable protein side chains buffer H^+ .
- Deoxyhemoglobin binds H^+ more readily than oxyhemoglobin.
- Respiratory compensation for metabolic acidosis lowers $PaCO_2$.

Application: Kussmaul breathing in ketoacidosis lowers CO_2 and partially raises pH. Carbonic anhydrase and the chloride shift support CO_2 transport by red cells.

Sources: Water and Buffer 1404.ppt, slides 29-35 (physiological buffers). Marks water/buffer reference, local reference pages 6-8 (bicarbonate, hemoglobin, chloride shift, and urinary buffering).

Past-paper checkpoints and wording traps

A buffer solution is 0.080 M in lactic acid ($pK_a=3.74$) and 0.070 M in sodium lactate. The pH of the solution is?

3.68 is the best answer. Henderson-Hasselbalch gives $3.74 + \log(0.070/0.080)$ 3.68.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q31

The chemical reactions in the small intestine take place in a basic (alkaline) environment. Which number indicates this basic pH?

8.5 is the best answer. pH 8.5 is clearly alkaline; values below 7 are acidic.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q34

For octanoic acid, $pK_a = 4.89$. Which pH range would be the best buffering region for this acid?

3.89-5.89 is the best answer. A weak acid buffers effectively within about one pH unit of its pK_a .

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q40

Hydroxyapatite has a very small dissolution equilibrium constant (approximately 10^{-26} in the source). Which description is correct under physiological conditions?

A very small dissolution equilibrium constant favors solid hydroxyapatite. Acid promotes dissolution; added calcium/phosphate can suppress it by the common-ion effect. Source correction: hydroxyapatite is the inorganic mineral component of bone, not the organic matrix.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q51

As the pKa of an acid increase, the acid will be:

Weaker is the best answer. A larger pKa means a smaller Ka and therefore a weaker acid.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q27

Which of the following is not an example of a weak acid?

Sulfuric acid is the best answer. Sulfuric acid is treated as a strong acid, whereas lactic, pyruvic, and carbonic acids are weak acids.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q54

The strength of an organic acid depends on:

Electronegativity is the best answer. Electron-withdrawing electronegativity stabilizes the conjugate base and increases organic-acid strength.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q59

Polyprotic acids such as H₃PO₄ can act as acid-base buffers:

At pH values around any of their pKa's is the best answer. Each ionizable proton provides a separate buffer region centered near its own pKa.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q21

A buffer solution contains propionic acid and its conjugate base; the pKa of propionic acid is 4.88. At what pH would the solution buffer best?

5.0 is the best answer. A buffer works best near its pKa; pH 5.0 is closest to pKa 4.88.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q10

03 Amino acids

Source basis: Lippincott, Chapter 1; concept-specific teacher references below.

Common amino-acid structure and side-chain classification

A standard alpha-amino acid has an alpha carbon attached to an amino group, a carboxyl group, a hydrogen, and a distinctive side chain. The side chain determines most residue-specific chemistry and supports the practical classification into nonpolar, uncharged polar, acidic, and basic groups. At physiological pH, the alpha amino and carboxyl groups are usually protonated and deprotonated respectively.

- The R group distinguishes one amino acid from another.
- Nonpolar side chains favor hydrophobic interactions.
- Uncharged polar side chains can hydrogen bond.
- Acidic side chains are usually negative and basic side chains often positive at physiological pH.
- Proline has a secondary amino group; glycine has two alpha-carbon hydrogens.

Application: A single side-chain substitution can alter folding, solubility, charge, or catalytic activity.

Sources: Amino acids and Proteins.Part1.pdf, pages 7-15 (structure and side-chain classes). Lippincott 6e, Chapter 1, local extraction items 11 and 135.

Nonpolar residues and the hydrophobic effect

Nonpolar side chains avoid favorable contact with water and therefore cluster in the interior of soluble globular proteins. In membrane proteins the orientation can reverse, with hydrophobic residues exposed to lipid. This hydrophobic effect is a major force driving folding and can become pathological when a mutation creates an exposed hydrophobic patch.

- Nonpolar residues usually do not form ionic or hydrogen bonds through their side chains.
- They pack inside soluble proteins but can face membrane lipids.
- Hydrophobic clustering stabilizes a protein's three-dimensional structure.

Application: The beta-globin Glu-to-Val substitution in HbS creates a hydrophobic interaction that promotes deoxyhemoglobin polymerization.

Sources: Amino acids and Proteins.Part1.pdf, pages 10-15 (amino-acid side-chain classes). Lippincott 6e, Chapter 1, local extraction items 20 and 23.

Polar residues, phosphorylation, and disulfide bonds

Serine, threonine, and tyrosine contain hydroxyl groups that form hydrogen bonds and can be phosphorylated. Cysteine contains a sulfhydryl group; oxidation of two cysteines forms a covalent disulfide bond and the paired product is cystine. These modifications can alter protein activity, localization, or structural stability.

- Serine, threonine, and tyrosine are common phosphorylation sites.
- Asparagine and glutamine have polar amide side chains.
- Two cysteine thiols can oxidize to a disulfide.
- Disulfides are especially common in extracellular and secreted proteins.

Application: Kinases regulate many enzymes and receptors through reversible phosphorylation. The oxidizing extracellular environment favors disulfide stabilization.

Sources: Amino acids and Proteins.Part1.pdf, pages 12-13 and 31-33. Lippincott 6e, Chapter 1, local extraction item 35.

Acidic and basic side chains, especially histidine

Aspartate and glutamate carry negatively charged carboxylate side chains at physiological pH, whereas lysine and arginine are usually positively charged. Histidine's imidazole has a pKa near the physiological range, so local protein environments can switch it between protonated and neutral states. That makes histidine particularly useful for acid-base catalysis and buffering.

- Aspartate and glutamate are acidic and usually negative.
- Lysine and arginine are basic and usually positive.
- Histidine can donate or accept protons near physiological pH.
- Local electrostatic environment can shift a side-chain pKa.

Application: Histidines contribute to hemoglobin buffering and the Bohr effect. Catalytic histidine often acts as a proton shuttle in enzyme active sites.

Sources: Amino acids and Proteins.Part1.pdf, pages 14-15 (acidic/basic residues and histidine).

Amino-acid stereochemistry and the glycine exception

Except for glycine, standard amino acids have four different substituents on the alpha carbon and are chiral. Human ribosomal proteins are synthesized from L-amino acids. Glycine is achiral because its side chain is hydrogen, leaving two hydrogens attached to the alpha carbon.

- Chiral amino acids occur as D and L enantiomers.
- Human proteins use L-amino acids.
- Glycine is achiral.
- D-amino acids occur in some bacterial cell walls and antibiotics.

Application: Stereochemistry affects enzyme recognition and drug design.

Sources: Amino acids and Proteins.Part1.pdf, page 20 (glycine and L-amino acids). Lippincott 6e, Chapter 1 stereochemistry section.

Amino-acid ionization, titration, and isoelectric point

Amino acids are amphoteric and can carry positive, zero, or negative net charge as pH changes. The isoelectric point is the pH at which the average net charge is zero; below pI a molecule is more protonated and positive, while above pI it is more deprotonated and negative. Buffering is greatest near each ionizable group's pKa, not specifically at pI unless a pKa coincides with it.

- At pI, average net charge is zero.
- pH below pI gives a net positive charge.
- pH above pI gives a net negative charge.
- Titration curves reveal pKa values and buffering regions.
- Ionizable side chains add extra titration steps.

Application: Electrophoresis separates proteins according to net charge at the buffer pH. Solubility is often lowest near a protein's pI.

Sources: Amino acids and Proteins.Part1.pdf, pages 25-26 (isoelectric point). Lippincott 6e, Chapter 1, local PDF pages 10-29.

Aromatic amino acids and analytical recognition

Phenylalanine, tyrosine, and tryptophan contain aromatic rings, with tryptophan contributing most strongly to protein absorbance near 280 nm and tyrosine contributing substantially. Tyrosine also contains a phenolic hydroxyl that can hydrogen bond, be phosphorylated, and ionize at alkaline pH. Aromatic-ring nitration underlies the xanthoproteic protein test.

- Tryptophan is the strongest common contributor to A280.
- Tyrosine also absorbs near 280 nm; phenylalanine contributes less.
- Tyrosine is aromatic yet has a polar hydroxyl group.
- The xanthoproteic reaction detects aromatic residues by a yellow-to-orange color.

Application: A280 is commonly used to estimate protein concentration, but nucleic acids and other chromophores can interfere. Tyrosine phosphorylation is a central signaling mechanism.

Sources: Amino acids and Proteins.Part1.pdf, page 21 (aromatic amino acids and absorbance). Confirmed TUMS practical curriculum: xanthoproteic test.

Past-paper checkpoints and wording traps

Which of the following statements about solutions of amino acids at physiologic pH is true?

All amino acids contain both positive and negative charges is the best answer. At physiological pH, free amino acids generally contain both NH_3^+ and COO^- groups as zwitterions.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q30

Choose the incorrect statement about amino acids.

D-amino acids occur, for example, in bacterial peptidoglycan and as D-serine in humans. Most ribosomally incorporated amino acids are L, with achiral glycine. The vague source phrase “tyrosine is a modified amino acid” is clarified here.

Source item: Biochemistry 1 · Finals 2022; Biochemistry · original Q19

An amino acid not found in proteins is ...

β -Alanine is the best answer. β -Alanine is not one of the standard α -amino acids incorporated into proteins.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q36

04 Protein structure and folding

Source basis: Lippincott, Chapter 2; concept-specific teacher references below.

Primary structure encodes folding and function

Primary structure is the linear amino-acid sequence of a polypeptide, conventionally read from N terminus to C terminus. Sequence determines the pattern of side-chain interactions available during folding and therefore strongly constrains final structure and function. A sequence-changing mutation can cause loss of function, instability, mislocalization, or toxic aggregation.

- Primary structure is amino-acid sequence.
- Sequences are written N terminus to C terminus.
- The sequence contains the information needed for native folding, although cells often require chaperones.
- A changed sequence can produce a changed phenotype.

Application: Missense variants can cause sickle cell disease or destabilize collagen. Sequence comparison helps interpret genetic disease variants.

Sources: Amino acids and Proteins.Part1.pdf, pages 31-33 and 43-49. Lippincott 6e, Chapter 2, local extraction items 5-6.

Peptide-bond formation, resonance, and planarity

A peptide bond is an amide linkage formed between an alpha-carboxyl group and an alpha-amino group. Resonance gives the C-N bond partial double-bond character, making the peptide unit rigid and planar and restricting rotation. Rotation around bonds adjacent to the alpha carbon still permits organized folding, and peptide bonds are usually trans.

- Peptide formation is a condensation reaction.
- The peptide C-N bond has partial double-bond character.
- Restricted rotation makes the peptide unit planar.
- Most peptide bonds adopt the trans configuration.
- Denaturation usually does not hydrolyze peptide bonds.

Application: Proteases hydrolyze selected peptide bonds during digestion, signaling, and zymogen activation.

Sources: Amino acids and Proteins.Part1.pdf, pages 31-33 (peptide bond). Lippincott 6e, Chapter 2, local PDF pages 30-54.

Alpha helices, beta sheets, and beta turns

Secondary structure is local, repetitive backbone folding stabilized mainly by hydrogen bonds between peptide carbonyl and amide groups. In an alpha helix, hydrogen bonds form within one chain and the side chains project outward; beta sheets contain extended strands hydrogen-bonded in parallel or antiparallel arrangements. Beta turns reverse chain direction and often connect nearby antiparallel strands.

- An alpha helix has about 3.6 residues per turn and i-to-i+4 backbone hydrogen bonds.
- Proline often disrupts an alpha helix.
- Beta sheets are stabilized by backbone hydrogen bonds between extended strands.
- Beta turns permit compact folding and often contain glycine or proline.

Application: Amyloid fibrils are enriched in ordered beta-sheet structure.

Sources: Amino acids and Proteins.Part1.pdf, pages 36-40 (alpha helix, beta sheet, and beta turn).

Tertiary structure, motifs, domains, and side-chain forces

Tertiary structure is the overall three-dimensional arrangement of one polypeptide, including how secondary elements assemble into motifs and domains. A domain is a compact structural and often functional unit that can fold somewhat independently. Hydrophobic interactions, hydrogen bonds, ionic interactions, and disulfide bonds collectively stabilize the native conformation.

- Motifs are recurring combinations of secondary structures.
- Domains are modular structural and functional units.
- Hydrophobic burial is a major folding force in aqueous solution.
- Ionic and hydrogen bonds are sensitive to pH and solvent.
- Disulfides provide covalent stabilization, especially extracellularly.

Application: A domain-specific mutation can selectively abolish binding or catalysis while leaving other domains intact.

Sources: Amino acids and Proteins.Part1.pdf, page 43 (protein domains). Lippincott 6e, Chapter 2, local extraction item 91.

Quaternary structure requires multiple subunits

Quaternary structure describes the arrangement and interaction of two or more polypeptide subunits. A one-chain protein can have tertiary structure but has no quaternary structure. Subunits are held mainly by noncovalent forces, although occasional interchain disulfides can contribute, and subunit communication permits cooperativity and allostery.

- One polypeptide has no quaternary structure.
- Two or more interacting polypeptides create quaternary structure.
- Hydrophobic, ionic, and hydrogen-bond interactions usually dominate subunit association.
- Hemoglobin is an $\alpha_2\beta_2$ cooperative tetramer.

Application: A variant at a subunit interface can alter cooperativity without directly changing an active or ligand-binding site.

Sources: Amino acids and Proteins.Part1.pdf, page 49 (quaternary structure). Lippincott 6e, Chapter 2, local extraction item 160.

Protein folding and molecular chaperones

Amino-acid sequence drives folding toward a low-energy native conformation, with the hydrophobic effect as a major early force. Molecular chaperones bind exposed hydrophobic regions, prevent inappropriate aggregation, and often use ATP to give a protein another opportunity to fold correctly. Chaperones assist folding but do not dictate a different final sequence-encoded structure.

- Folding is ordered rather than a random search of all conformations.
- Hydrophobic collapse helps organize early folding intermediates.
- Hsp70 protects exposed hydrophobic regions of nascent chains.
- Chaperonins provide a protected folding chamber.
- Many chaperone cycles require ATP.

Application: Cell stress increases heat-shock proteins. Failure of proteostasis contributes to neurodegeneration and some loss-of-function diseases.

Sources: Exam scope update 2026-08-20: complete confirmed Lippincott scaffold includes Chapter 2. Lippincott 6e, Chapter 2, local extraction items 101, 110, and 115.

Denaturation disrupts higher structure, not primary sequence

Denaturation unfolds and disorganizes secondary and tertiary structure without ordinarily hydrolyzing peptide bonds, so primary structure remains intact. Heat, extreme pH, organic solvents, detergents, urea, and heavy metals can disrupt the interactions maintaining the native state. Some proteins refold after the denaturant is removed, but aggregation makes denaturation commonly irreversible in practice.

- Denaturation causes loss of native conformation and usually function.
- Peptide bonds are generally preserved.
- Hydrophobic exposure can cause precipitation and aggregation.
- Reversibility depends on sequence, conditions, and avoidance of aggregation.

Application: Fever and severe acidosis can impair enzyme function, although ordinary fever is not equivalent to wholesale protein denaturation. Heat inactivates many enzymes used in laboratory samples.

Sources: Amino acids and Proteins.Part1.pdf, page 48 (protein denaturation). Lippincott 6e, Chapter 2, local extraction item 107.

Protein misfolding, amyloid, and prion propagation

Misfolded proteins are normally refolded or targeted for degradation, but failed quality control can allow toxic aggregates to accumulate. Amyloid is an ordered, insoluble, beta-sheet-rich fibrillar assembly rather than a random denatured precipitate. In prion disease, a pathogenic PrP conformation templates conversion of normal host PrP into the same abnormal conformation.

- Amyloid fibrils are ordered and beta-sheet rich.
- Alzheimer plaques contain A beta fragments produced from APP.
- PrPSc promotes conformational conversion of PrPC.
- Sequence mutation is not required for every misfolding disease.

Application: Amyloid disorders include Alzheimer disease and systemic amyloidoses. Prion diseases are transmissible spongiform encephalopathies and resist ordinary proteolysis.

Sources: Exam scope update 2026-08-20: complete confirmed Lippincott scaffold includes Chapter 2. Lippincott 6e, Chapter 2, local extraction items 125, 148, and 172.

Past-paper checkpoints and wording traps

A tripeptide has?

3 amino acids and 2 peptide bonds is the best answer. Three amino acids are linked by two peptide bonds.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q29

Peptide bond is a?

Covalent bond is the best answer. A peptide bond is a covalent amide bond.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q39

The oxygen carrier of muscle is the globular protein myoglobin. Which one of the following amino acids is highly likely to be localized within the interior of the molecule?

Valine is the best answer. Valine is nonpolar and is therefore favored in the hydrophobic interior of a globular protein.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q13

A coiled peptide chain held in place by hydrogen bonding between peptide bonds in the same chain (not between two chains) is?

α -helix is the best answer. An α -helix is stabilized by intrachain backbone hydrogen bonds.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q68

The secondary structure of protein is held together by which type of bonding?

Hydrogen bonding is the best answer. Backbone hydrogen bonds stabilize α -helices and β -sheets.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q34

05 Hemoglobin and globular proteins

Source basis: Lippincott, Chapter 3; concept-specific teacher references below.

Heme chemistry and HbA architecture

Heme consists of protoporphyrin IX coordinated to ferrous iron, Fe^{2+} , which can bind oxygen reversibly. Major adult hemoglobin HbA is an $\alpha_2\beta_2$ tetramer with one heme per subunit and therefore four oxygen-binding sites. Oxidation of iron to Fe^{3+} produces methemoglobin, which cannot bind oxygen normally.

- Functional heme oxygen binding requires Fe^{2+} .
- HbA contains two alpha and two beta chains.
- Each globin subunit contains one heme.
- Hemoglobin subunits associate mainly through noncovalent forces.

Application: Oxidant drugs can convert Fe^{2+} to Fe^{3+} and cause methemoglobinemia.

Sources: exam-scope.md, August 25 active biochemistry scope: proteins and hemoglobin; complete Lippincott Chapters 1-7 scaffold. Lippincott 6e, Chapter 3, local extraction items 8 and 28.

Myoglobin stores oxygen; hemoglobin transports it

Myoglobin is a monomeric muscle hemeprotein that stores and facilitates intracellular oxygen movement, whereas hemoglobin is a tetrameric red-cell transporter. Myoglobin binds one oxygen and has a hyperbolic, high-affinity curve without cooperativity. Hemoglobin binds up to four oxygen molecules and uses subunit interactions to load in lungs and unload in tissues.

- Myoglobin is monomeric and found in cardiac and skeletal muscle.
- Hemoglobin is tetrameric and found in erythrocytes.
- Myoglobin has a hyperbolic oxygen-binding curve.
- Hemoglobin has a sigmoidal oxygen-binding curve.

Application: Myoglobin rises early after muscle injury but is less cardiac-specific than troponin.

Sources: Lippincott 6e, Chapter 3, local extraction items 7, 12-16, and 41-53. exam-scope.md, complete confirmed Lippincott Chapters 1-7 scaffold.

Cooperativity, T/R states, and P50

Deoxyhemoglobin favors the low-affinity T state, whereas oxygen binding shifts subunits toward the high-affinity R state. Binding at one heme increases affinity at the remaining hemes, producing the sigmoidal curve and efficient tissue unloading. P50 is the oxygen partial pressure at half-saturation; a higher P50 means lower affinity and a right shift.

- T state is tense, deoxygenated, and lower affinity.
- R state is relaxed, oxygenated, and higher affinity.
- Positive cooperativity creates a sigmoidal curve.
- Higher P50 means lower oxygen affinity.

Application: Right shifts improve unloading but can impair loading if extreme; left shifts do the reverse.

Sources: Lippincott 6e, Chapter 3, local extraction items 39-40, 49, and 53-64.

Bohr effect: active tissues promote oxygen unloading

In metabolically active tissues, increased H^+ and CO_2 stabilize deoxyhemoglobin's T state and lower oxygen affinity, shifting the curve right. In the lungs, CO_2 removal and higher pH favor oxygen loading. CO_2 also forms carbamates at globin N termini, while deoxyhemoglobin helps buffer protons.

- Lower pH decreases hemoglobin oxygen affinity.
- Higher CO₂ favors oxygen unloading.
- The effect is reversed in the lungs.
- Deoxyhemoglobin binds H⁺ more strongly than oxyhemoglobin.

Application: Exercise increases local CO₂, H⁺, and temperature, promoting oxygen delivery. Acidosis can worsen HbS deoxygenation and sickling.

Sources: Water and Buffer 1404.ppt, slides 33-35 (hemoglobin buffering and CO₂ transport). Lippincott 6e, Chapter 3, local extraction item 65.

2,3-BPG stabilizes deoxyhemoglobin

2,3-BPG is an anionic red-cell metabolite that binds the positively charged central cavity of deoxyhemoglobin. It stabilizes the T state, lowers oxygen affinity, shifts the curve right, and thereby enhances tissue unloading. Its concentration rises in chronic hypoxia and anemia, whereas stored blood loses 2,3-BPG and initially unloads oxygen less effectively.

- One 2,3-BPG binds per deoxyhemoglobin tetramer.
- It binds chiefly between the beta chains.
- It lowers oxygen affinity and raises P₅₀.
- Hypoxia and chronic anemia increase its concentration.

Application: High-altitude adaptation includes increased 2,3-BPG. Freshly transfused stored red cells may temporarily have reduced unloading capacity.

Sources: Lippincott 6e, Chapter 3, local extraction items 77-91.

Fetal hemoglobin and placental oxygen transfer

Fetal hemoglobin is $\alpha_2\gamma_2$. Gamma chains bind 2,3-BPG less strongly than adult beta chains, so HbF has higher oxygen affinity than HbA in the same red-cell environment. This left shift enables fetal blood to acquire oxygen from maternal blood across the placenta.

- HbF composition is $\alpha_2\gamma_2$.
- HbF binds 2,3-BPG weakly.
- Weaker BPG binding increases oxygen affinity.
- HbF predominates before birth and falls after birth.

Application: Increasing HbF with hydroxyurea reduces HbS polymerization and sickle crises.

Sources: Lippincott 6e, Chapter 3, local extraction items 108 and 116-119.

Carbon monoxide poisoning versus methemoglobinemia

Carbon monoxide occupies heme sites and stabilizes the R state, reducing oxygen capacity while making remaining sites hold oxygen more tightly. Methemoglobin instead contains Fe³⁺, which cannot bind oxygen normally and can produce brown blood with cyanosis. Both cause tissue hypoxia, but pulse oximetry and treatment considerations differ.

- CO reduces available sites and left-shifts remaining oxygen binding.
- Methemoglobin contains Fe³⁺ rather than Fe²⁺.
- Chocolate-brown blood suggests methemoglobinemia.
- Oxidant drugs such as nitrates or benzocaine can cause acquired methemoglobinemia.

Application: CO poisoning is treated with high-concentration oxygen and sometimes hyperbaric oxygen. Methylene blue can reduce methemoglobin in appropriate patients but requires clinical caution.

Sources: Exam scope update 2026-08-20: complete confirmed Lippincott scaffold includes Chapter 3. Lippincott 6e, Chapter 3, local extraction items 97-104 and 167-174.

HbA1c records cumulative glucose exposure

HbA1c forms by slow nonenzymatic glycation of HbA, especially at the N-terminal valines of beta chains. Its abundance reflects average glucose exposure over the circulating red cell population rather than a momentary glucose value. Altered red-cell lifespan, transfusion, or some hemoglobin variants can make HbA1c misleading.

- Glycation is nonenzymatic and differs from enzyme-directed glycosylation.
- The extent of HbA1c formation rises with ambient glucose concentration and time.
- HbA1c reflects roughly the preceding two to three months with greater weighting of recent weeks.
- Conditions changing erythrocyte lifespan can bias the result.

Application: HbA1c is used to diagnose and monitor diabetes when the assay is reliable. Recent blood loss or hemolysis can produce an unexpectedly low HbA1c.

Sources: Exam scope update 2026-08-20: complete confirmed Lippincott scaffold includes Chapter 3. Lippincott 6e, Chapter 3, local extraction items 122-128 and 204.

HbS: a primary-structure change causes polymerization

HbS contains a beta-chain Glu6Val substitution that creates a hydrophobic patch. When deoxygenated, HbS molecules polymerize into rigid fibers that distort red cells, impair deformability, and occlude small vessels. Low oxygen, acidosis, dehydration, and other conditions favoring deoxyhemoglobin increase sickling, whereas HbF reduces polymerization.

- The mutation is beta6 glutamate to valine.
- Deoxy-HbS, not oxy-HbS, polymerizes most readily.
- Polymerization causes hemolysis and vaso-occlusion.
- Hydroxyurea helps by increasing HbF.

Application: Vaso-occlusion causes painful crises, infarction, and organ damage. Newborn screening enables prophylaxis before severe infection occurs.

Sources: Lippincott 6e, Chapter 1 local extraction item 24 and Chapter 3 local extraction items 151-158.

Thalassemia is impaired globin-chain production

Thalassemias result from decreased or absent synthesis of alpha- or beta-globin chains rather than a single abnormal hemoglobin structure alone. The unbalanced normally synthesized partner chains precipitate, damage erythroid precursors, and cause ineffective erythropoiesis and hemolysis. Alpha-thalassemia is commonly deletional, while beta-thalassemia commonly reflects point mutations affecting expression or RNA processing.

- Thalassemia is primarily a quantitative globin defect.
- Beta-thalassemia leaves excess alpha chains that precipitate.
- Severity depends on how many genes are affected and whether production is reduced or absent.
- Beta-thalassemia often raises HbA2 and HbF.

Application: Severe disease produces anemia, marrow expansion, transfusion dependence, and iron overload. Microcytosis out of proportion to anemia can suggest thalassemia trait.

Sources: Exam scope update 2026-08-20: complete confirmed Lippincott scaffold includes Chapter 3. Lippincott 6e, Chapter 3, local extraction items 175-193.

Past-paper checkpoints and wording traps

Which of the following mutations would produce a minor thalassemia?

Deletion of one β -globin locus is the best answer. Loss of one β -globin locus can produce β -thalassemia trait/minor.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q32

The ability of hemoglobin to serve as an effective transporter of oxygen and carbon dioxide between lungs and tissues is explained by which of the following properties?

In contrast to myoglobin, hemoglobin exhibits greater changes in secondary and tertiary structure after oxygen binding is the best answer. Oxygen binding produces cooperative conformational changes in hemoglobin, unlike monomeric myoglobin.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q83

Which of the following would produce Methemoglobinemia?

Converting Ferrous iron (Fe^{2+}) to Ferric (Fe^{3+}) is the best answer. Oxidation of heme iron from Fe^{2+} to Fe^{3+} forms methemoglobin, which cannot bind oxygen normally.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q47

Blood is drawn from a child with severe anemia and the hemoglobin protein is degraded for the analysis of peptide and amino acids. Of the results below, which change in hemoglobin primary structure is most likely to correlate with the clinical phenotype of anemia?

Leu-Glu-Ile to Leu-Val-Ile is the best answer. Replacing glutamate with valine reproduces the key nonconservative change of sickle hemoglobin.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q57

All of the following statements correctly describes transport of O_2 by hemoglobin except

Increased CO_2 concentrations increase O_2 affinity is the best answer. Higher CO_2 lowers hemoglobin oxygen affinity through the Bohr effect rather than increasing it.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q51

In sickle cell anemia, the basis of the malfunction of the hemoglobin molecule is ...

Substitution of a single amino acid is the best answer. Sickle hemoglobin results from a single β -globin substitution, Glu6Val, changing primary structure.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q20

06 Collagen, elastin and fibrous proteins

Source basis: Lippincott, Chapter 4; concept-specific teacher references below.

Collagen triple helix, Gly-X-Y, and major types

Collagen is a triple helix of three alpha chains with a repeating Gly-X-Y sequence, commonly containing proline and hydroxyproline. Glycine must occur every third residue because only its small hydrogen side chain fits at the helix center. Types I, II, and III form fibrils, whereas type IV forms a sheet-like basement-membrane network.

- Collagen is the most abundant protein in the body.
- Gly-X-Y repeats permit triple-helix packing.
- Type I predominates in bone, skin, tendon, and scar.
- Type II predominates in cartilage and vitreous body.
- Type III forms reticular fibers; type IV forms basement-membrane networks.

Application: Tissue distribution helps predict the phenotype of a collagen defect.

Sources: exam-scope.md, complete confirmed Lippincott Chapters 1-7 scaffold. Lippincott 6e, Chapter 4, local PDF pages 87-104.

Intracellular collagen synthesis and vitamin C

Prepro-alpha chains enter rough ER, lose their signal sequence, and undergo selected proline and lysine hydroxylation plus hydroxylysine glycosylation. Hydroxylases require oxygen, Fe²⁺, and vitamin C; terminal propeptides align three chains and prevent premature fibril formation. The chains assemble from the C terminus into a procollagen triple helix before Golgi processing and secretion.

- Hydroxylation occurs in rough ER after residues enter the chain.
- Vitamin C keeps iron in the reduced state needed by hydroxylases.
- Hydroxyproline stabilizes the triple helix through hydrogen bonding.
- Propeptides align chains and keep procollagen soluble inside the cell.

Application: Vitamin C deficiency causes weak collagen, bleeding, and poor wound healing.

Sources: Lippincott 6e, Chapter 4, local PDF pages 87-104 and extraction items 31-39.

Extracellular procollagen cleavage and copper-dependent cross-linking

After secretion, N- and C-procollagen peptidases remove terminal propeptides to produce tropocollagen. Tropocollagen molecules align into staggered fibrils, and copper-dependent lysyl oxidase creates reactive aldehydes from lysine or hydroxylysine. Covalent intermolecular cross-links then provide tensile strength.

- Procollagen cleavage occurs extracellularly.
- Tropocollagen self-assembles into staggered fibrils.
- Lysyl oxidase requires copper.
- Cross-link density increases tensile strength with maturation.

Application: Copper deficiency or lysyl-oxidase inhibition weakens collagen cross-linking. Defective procollagen peptidases can cause Ehlers-Danlos phenotypes.

Sources: Lippincott 6e, Chapter 4, local PDF pages 87-104 and extraction items 40-47.

Collagen degradation and matrix remodeling

Collagen is long lived but not permanent; collagenases initiate cleavage of the otherwise resistant triple helix. These enzymes belong to the matrix metalloproteinase family and produce fragments that other proteases further degrade. Synthesis and degradation must remain balanced during development, wound healing, and tissue maintenance.

- Collagenases are matrix metalloproteinases.
- Initial collagenase cleavage enables further proteolysis.
- MMP activity is regulated by activation and tissue inhibitors.
- Excess breakdown weakens matrix; excess deposition causes fibrosis.

Application: MMP activity contributes to arthritis, tumor invasion, and vascular remodeling. Chronic injury can shift balance toward fibrotic collagen deposition.

Sources: Exam scope update 2026-08-20: complete confirmed Lippincott scaffold includes Chapter 4. Lippincott 6e, Chapter 4, local extraction items 52-53.

Scurvy, osteogenesis imperfecta, and Ehlers-Danlos syndrome

Scurvy is acquired defective hydroxylation from vitamin C deficiency, causing capillary fragility and poor wound healing. Osteogenesis imperfecta usually involves type I collagen quantity or structure and produces fractures, blue sclerae, and sometimes hearing or dental abnormalities. Ehlers-Danlos syndromes arise from varied collagen or processing defects and commonly feature skin hyperextensibility and joint hypermobility; vascular type III collagen defects risk arterial rupture.

- Scurvy impairs proline and lysine hydroxylation.
- Osteogenesis imperfecta commonly involves type I collagen.
- Glycine substitution can disrupt the entire triple helix.
- Classic EDS often involves type V; vascular EDS involves type III.
- Dominant-negative chains can damage a multichain collagen molecule.

Application: Bleeding plus dietary deficiency suggests scurvy; congenital recurrent fractures suggest OI. Marked hypermobility with fragile hyperextensible skin suggests EDS.

Sources: Lippincott 6e, Chapter 4, local PDF pages 87-104 and extraction item 58.

Elastin structure permits stretch and recoil

Collagen resists stretch, whereas elastin permits tissues to extend and recoil. Tropoelastin is rich in small nonpolar residues and lysine and is deposited on a fibrillin microfibril scaffold. Lysyl oxidase contributes to desmosine cross-links, creating an extensible, amorphous network important in lungs, large arteries, and elastic ligaments.

- Elastin provides elasticity rather than high tensile strength.
- Tropoelastin is the soluble precursor.
- Fibrillin microfibrils guide elastic-fiber assembly.
- Desmosine cross-links connect tropoelastin chains.
- Elastic recoil is crucial in alveoli and large arteries.

Application: Loss of alveolar elastin reduces recoil and promotes expiratory airway collapse.

Sources: Lippincott 6e, Chapter 4, local extraction items 68-74.

Fibrillin defects and unopposed elastase injury

Fibrillin-1 forms the microfibril scaffold for elastic fibers, and its mutation causes Marfan syndrome with skeletal, ocular, and cardiovascular manifestations. Alpha1-antitrypsin normally restrains neutrophil elastase in the lung; deficiency permits destruction of alveolar elastic tissue and panacinar emphysema. The common severe Z variant also polymerizes in hepatocyte ER, producing both low circulating inhibitor and liver injury.

- Fibrillin-1 mutation causes Marfan syndrome.
- Alpha1-antitrypsin inhibits neutrophil elastase.

- Unopposed elastase destroys alveolar walls and recoil.
- Misfolded Z-variant AAT can accumulate in hepatocytes.

Application: Marfan syndrome raises risk of aortic root disease and lens dislocation. AAT deficiency causes early panacinar emphysema, worsened by smoking, and may cause cirrhosis.

Sources: Exam scope update 2026-08-20: complete confirmed Lippincott scaffold includes Chapter 4. Lippincott 6e, Chapter 4, local extraction items 74 and 78-85.

Past-paper checkpoints and wording traps

All of the following statements regarding fibrous proteins are true except

Hemoglobin is the best example is the best answer. Hemoglobin is globular, whereas collagen and keratin are fibrous structural proteins.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q33

Which of the substances below is primarily found in tendons?

Collagen is the best answer. Protein function follows primary sequence and higher-order folding interactions.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q49

During synthesis of mature collagen fiber, which one of the following steps would occur within the fibroblast?

Formation of a triple helix is the best answer. Protein function follows primary sequence and higher-order folding interactions.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q43

Deficiency of lysyl hydroxylase causes syndrome characterized by hypermobile joints and skin abnormalities.

Ehlers-Danlos is the best answer. Protein function follows primary sequence and higher-order folding interactions.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q80

A child with tall stature, loose joints, and detached retinas is found to have a mutation in type II collagen. Recall that collagen consists of a repeating tripeptide motif where the first amino acid of each tripeptide is the same. Which of the following amino acids is the recurring amino acid most likely to be altered in mutations that distort collagen molecules?

Glycine is the best answer. Protein function follows primary sequence and higher-order folding interactions.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q11

The highest concentration of cystine can be found in

Keratin is the best answer. Keratin is rich in cystine disulfide bonds, which strengthen hair and nails.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q27

Which one of the following results in Ehlers-Danlos syndrome?

Deficiency of lysyl hydroxylase is the best answer. Lysyl hydroxylase defects impair collagen maturation and can cause hyperextensible skin and hypermobile joints.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q32

Glycine and proline are the most abundant amino acids in the structure of

Collagen is the best answer. Collagen is rich in glycine and proline/hydroxyproline, supporting its triple helix.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q44

07 Enzymes, kinetics and regulation

Source basis: Lippincott, Chapter 5; concept-specific teacher references below.

Enzyme kinetics at a glance

Pattern	K_m	V_{max} / distinguishing point
Michaelis-Menten	$[S]$ at $v = V_{max}/2$	$v = V_{max}[S]/(K_m + [S])$; at $[S] = 3K_m$, $v = 0.75 V_{max}$.
More enzyme	Unchanged	V_{max} rises proportionally when other conditions are fixed.
Competitive inhibition	Increased apparent K_m	V_{max} unchanged; high substrate can overcome reversible competition.
Pure noncompetitive inhibition	Unchanged	V_{max} decreased. Mixed inhibition need not leave K_m unchanged.
Uncompetitive inhibition	Decreased apparent K_m	V_{max} decreased in the same ratio; inhibitor binds ES.
Irreversible inactivation	No universal K_m prediction	Active enzyme falls; removing free inhibitor does not restore already-inactivated enzyme.
Catalytic performance	K_m alone is insufficient	$k_{cat} = V_{max}/[E]_{total}$; k_{cat}/K_m is the specificity constant at low $[S]$.

Active sites create specificity and induced fit

An enzyme active site is a three-dimensional pocket formed by residues that may be far apart in primary sequence. Complementary shape, charge, polarity, and dynamics give substrate and reaction specificity. Substrate binding can induce a conformational change that positions catalytic groups and stabilizes the transition state.

- The active site is created by tertiary structure.
- Specificity can be absolute, group, bond, or stereochemical.
- Induced fit is conformational adjustment after binding.
- Catalytic residues need not be adjacent in sequence.

Application: Many drugs work by occupying an enzyme active site or an allosteric pocket. Stereospecificity explains why drug enantiomers can differ in effect.

Sources: exam-scope.md, August 25 active biochemistry scope: enzymes and regulation; complete Lippincott Chapters 1-7 scaffold. Lippincott 6e, Chapter 5, local extraction items 12, 14, and 18.

Enzymes lower activation energy without changing equilibrium

Enzymes accelerate reactions by stabilizing the transition state and providing a lower-activation-energy pathway. They do not change the overall free-energy difference between reactants and products and do not move the equilibrium position. They accelerate both forward and reverse reactions, allowing equilibrium to be reached faster.

- Activation energy is the barrier to reaching the transition state.
- Enzymes lower activation energy.
- Enzymes do not change reaction delta G.

- Enzymes do not change the equilibrium constant.
- Transition-state stabilization is central to catalysis.

Application: Transition-state analogs can be potent enzyme inhibitors and useful drugs.

Sources: Lippincott 6e, Chapter 5, local extraction items 29-36.

Apoenzyme, holoenzyme, cofactor, and coenzyme

An apoenzyme is the protein portion without a required nonprotein component, whereas the complete active form is the holoenzyme. Metal ions can act as cofactors; organic helpers are coenzymes, many derived from vitamins. A cosubstrate binds transiently and leaves chemically changed, while a prosthetic group remains tightly bound and is regenerated on the enzyme.

- Apoenzyme plus required nonprotein component equals holoenzyme.
- Metal ions such as Zn^{2+} or Fe^{2+} can be cofactors.
- NAD^+ is a vitamin-derived coenzyme and cosubstrate.
- Tightly bound FAD can function as a prosthetic group.

Application: Vitamin deficiencies can reduce the activity of all enzymes requiring the corresponding coenzyme. Metal chelation can inhibit metalloenzymes.

Sources: Vitamin 2024.pdf, pages 4-12 (cofactors, coenzymes, and prosthetic groups). Lippincott 6e, Chapter 5, local extraction items 20-25 and 114.

Temperature and pH alter enzyme rate

Raising temperature initially increases molecular collisions and reaction rate, but excessive heat unfolds enzymes and reduces activity. pH changes the protonation of catalytic and substrate-binding residues, and extreme pH can disrupt the protein's stabilizing interactions. Each enzyme therefore has an activity profile suited to its physiological compartment.

- Rate rises with temperature until structural instability dominates.
- High heat decreases activity through denaturation.
- pH controls active-site residue ionization.
- Extreme pH can disrupt tertiary structure.
- Pepsin and cytosolic enzymes have different pH optima.

Application: Specimen handling temperature and pH can change measured enzyme activity. Acidic lysosomes support hydrolases that would be less active in neutral cytosol.

Sources: Lippincott 6e, Chapter 5, local extraction items 43-48.

Michaelis-Menten saturation, K_m , and V_{max}

For a simple Michaelis-Menten enzyme, velocity rises hyperbolically with substrate concentration and approaches V_{max} as active sites become saturated. K_m is the substrate concentration at half V_{max} and is often used as an inverse indicator of apparent substrate affinity. V_{max} depends on enzyme concentration, whereas K_m for a given enzyme-substrate system does not.

- $v = V_{max}[S]/(K_m + [S])$.
- At $[S] = K_m$, $v = V_{max}/2$.
- Lower K_m commonly means higher apparent affinity.
- At saturation, adding substrate does not raise velocity further.
- Increasing enzyme concentration increases V_{max} .

Application: K_m and V_{max} help distinguish inhibitor mechanisms and compare enzyme variants.

Sources: Enzyme Kinetics and regulation- Session2 2024.pptx, slides 7 and 9-10.

Lineweaver-Burk plot interpretation

The Lineweaver-Burk plot linearizes Michaelis-Menten kinetics by plotting $1/v$ against $1/[S]$. Its y-intercept is $1/V_{max}$, x-intercept is $-1/K_m$, and slope is K_m/V_{max} . Although modern fitting generally uses the original nonlinear equation, the plot remains a common exam tool for recognizing parameter changes.

- y-intercept = $1/V_{max}$.
- x-intercept = $-1/K_m$.
- slope = K_m/V_{max} .
- Competitive and noncompetitive inhibitors produce different intersection patterns.

Sources: Enzyme Kinetics and regulation- Session2 2024.pptx, slides 13-14. Lippincott 6e, Chapter 5, local extraction items 62-65.

Competitive inhibition raises apparent K_m

A competitive inhibitor and substrate compete for the enzyme's active site. It increases apparent K_m because more substrate is needed to reach half V_{max} , but V_{max} is unchanged because sufficiently high substrate can outcompete reversible inhibitor. On a Lineweaver-Burk plot, lines share the y-intercept.

- Competitive inhibitor binds the active site.
- Apparent K_m increases.
- V_{max} is unchanged.
- Increasing substrate can overcome reversible competitive inhibition.

Application: Fomepizole or ethanol competitively limits toxic alcohol metabolism by alcohol dehydrogenase. Statins competitively inhibit HMG-CoA reductase.

Sources: Enzyme Kinetics and regulation- Session2 2024.pptx, slides 16-18 and 21. Lippincott 6e, Chapter 5, local extraction item 112.

Noncompetitive versus irreversible inhibition

A pure noncompetitive inhibitor binds enzyme and enzyme-substrate complex outside the active site, lowering V_{max} without changing K_m . An irreversible inhibitor permanently inactivates enzyme, commonly through covalent modification, so activity returns only when new enzyme is synthesized. Neither mechanism is overcome simply by increasing substrate concentration.

- Pure noncompetitive inhibition decreases V_{max} and leaves K_m unchanged.
- Irreversible inhibition permanently reduces active enzyme concentration.
- Increasing substrate does not restore the original V_{max} .
- Real mixed inhibition can change both K_m and V_{max} .

Application: Aspirin irreversibly acetylates cyclooxygenase. Organophosphates irreversibly inhibit acetylcholinesterase unless treated before aging.

Sources: Enzyme Kinetics and regulation- Session2 2024.pptx, slides 19-24. Lippincott 6e, Chapter 5, local extraction item 82.

Allosteric enzymes control pathway flux

Allosteric effectors bind regulatory sites distinct from the active site and shift an enzyme between activity states. These enzymes are often multisubunit, show sigmoidal rather than simple Michaelis-Menten kinetics, and catalyze rate-limiting or committed steps. Feedback inhibition occurs when a pathway end product inhibits an earlier regulatory enzyme, preventing wasteful overproduction.

- Allosteric sites are distinct from catalytic sites.

- Allosteric enzymes often show cooperative sigmoidal kinetics.
- Regulatory enzymes commonly occur at irreversible pathway-control steps.
- End-product feedback conserves energy and precursors.

Application: Allosteric drugs can modulate activity without competing with substrate.

Sources: Enzyme Kinetics and regulation- Session2 2024.pptx, slides 35-38.

Fast and slow modes of enzyme regulation

Reversible phosphorylation rapidly changes existing enzyme activity and commonly targets serine, threonine, or tyrosine. Proteolytic cleavage activates zymogens irreversibly, which is useful when enzymes must remain off until they reach a specific site. Induction and repression alter enzyme abundance through synthesis or degradation and therefore act more slowly than allosteric or covalent control.

- Kinases add phosphate; phosphatases remove it.
- Phosphorylation can activate one enzyme and inhibit another.
- Zymogen activation is proteolytic and irreversible for that molecule.
- Induction increases enzyme synthesis over hours to days.
- Multiple regulation modes can act on the same pathway.

Application: Digestive proteases and coagulation factors are synthesized as zymogens. Inappropriate protease activation can cause pancreatitis or uncontrolled coagulation.

Sources: Enzyme Kinetics and regulation- Session2 2024.pptx, slides 39, 41, and 43.

Plasma enzymes, isoenzymes, and tissue injury

Many intracellular enzymes appear in plasma at low steady-state concentrations and rise when cells are injured. Isoenzymes catalyze the same reaction but differ in structure and tissue distribution, helping identify a source of damage. Biomarker interpretation requires tissue specificity and time course; for myocardial infarction, cardiac troponin has replaced CK-MB as the preferred marker in most settings.

- Plasma enzyme elevation can indicate cell injury rather than increased pathway activity in plasma.
- Isoenzymes differ in sequence, charge, kinetics, or tissue distribution.
- ALT is relatively useful for hepatocellular injury.
- Cardiac troponin is the preferred myocardial-injury biomarker.
- A result must be interpreted with symptoms, timing, and reference intervals.

Application: Serial troponin measurements help detect an evolving myocardial injury pattern. An enzyme pattern can support a tissue source but rarely establishes a diagnosis alone.

Sources: Exam scope update 2026-08-20: complete confirmed Lippincott scaffold includes Chapter 5. Lippincott 6e, Chapter 5, local extraction items 96-110.

Past-paper checkpoints and wording traps

Which of the following statements about Michaelis-Menten constant is correct?

A low K_m indicates a high affinity of an enzyme for its substrate. is the best answer. Michaelis-Menten parameters relate substrate concentration to reaction velocity. K_m is independent of total enzyme concentration in the Michaelis-Menten model. Low K_m is often used as an affinity shorthand, but K_m is not universally a dissociation constant. The original wording permits both A and the customary textbook interpretation of B.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q14

When $K_m = S$, reaction velocity (v) would be equal to.....

1/2 V_{max} is the best answer. Michaelis-Menten parameters relate substrate concentration to reaction velocity.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q43

Which of the following can act as a feedback inhibitor of an enzyme?

End product is the best answer. An end product commonly inhibits an early committed enzyme in its own biosynthetic pathway.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q53

Zymogen or proenzyme is a?

Enzyme precursor is the best answer. A zymogen is an inactive enzyme precursor activated by a later step such as proteolysis.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q65

The effect of reversible competitive inhibitor can be canceled out by?

Increasing the substrate concentration is the best answer. High substrate can outcompete a reversible competitive inhibitor for the active site.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q44

Which enzyme class catalyzes the reaction shown?

Lactate dehydrogenase couples lactate oxidation to NAD^+ reduction. It is an oxidoreductase.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q22

When $S=3K_m$, the V is equal to..... ?

3/4 V_{max} is the best answer. Michaelis-Menten parameters relate substrate concentration to reaction velocity.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q35

Ibuprofen is a reversible inhibitor of the COX-1 enzyme which is involved in Arachidonic Acid metabolism. Blocking of enzyme action by blocking its active site is called as?

Competitive inhibition is the best answer. Ibuprofen reversibly competes at the cyclooxygenase active site; aspirin inhibits COX irreversibly by acetylation.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q49

The catalytic efficiency of two enzymes acting on a substrate is commonly compared using which parameter?

Editorial repair: the source offered K_m alone. k_{cat}/K_m , not K_m alone, is the specificity constant used for low-substrate catalytic performance.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q8

Which one of the following is a reversible inhibitor of the COX-1 enzyme that is involved in prostaglandin synthesis?

Ibuprofen is the best answer. Ibuprofen reversibly competes at the cyclooxygenase active site; aspirin inhibits COX irreversibly by acetylation.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q53

A non-competitive inhibitor of an enzyme ...

Decreases V_{max} with no change in K_m is the best answer. Pure noncompetitive inhibition lowers V_{max} without changing K_m .

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q5

Which of the following can restore the activity of an enzyme already irreversibly inhibited?

The original four options cannot reverse irreversible enzyme inactivation. Adding E prevents teaching an incorrect source key.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q12

08 Bioenergetics and oxidative phosphorylation

Source basis: Lippincott, Chapter 6; concept-specific teacher references below.

Free energy, reaction direction, and energetic coupling

A negative actual free-energy change (ΔG) means a reaction is thermodynamically favorable in the written direction; $\Delta G = 0$ at equilibrium. Actual ΔG depends on concentrations, so it can differ from standard ΔG° . Cells drive an endergonic step by coupling it to a sufficiently exergonic process such as ATP hydrolysis, making the summed ΔG negative.

- ΔG predicts feasibility, not reaction rate
- Actual concentrations can change the sign of ΔG
- Free-energy changes of coupled reactions are additive
- ATP is kinetically stable but has high phosphate-transfer potential

Application: Energy failure during hypoxia ATP-dependent ion-pump failure

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 6, bioenergetics and free energy (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Electron-transport chain organization and electron flow

Electrons from NADH enter at Complex I and those from succinate-linked FADH₂ enter at Complex II; both routes converge on coenzyme Q, Complex III, cytochrome c, and Complex IV. Oxygen is the terminal electron acceptor and is reduced to water. Complexes I, III, and IV pump protons, whereas Complex II does not.

- ETC is in the inner mitochondrial membrane
- CoQ and cytochrome c are mobile carriers
- Complex II is also succinate dehydrogenase
- O₂ accepts electrons at Complex IV

Application: Cyanide and carbon-monoxide poisoning Mitochondrial respiratory-chain disease

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 6, electron transport (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Chemiosmosis, ATP synthase, and oxidative phosphorylation

Electron transport pumps protons from the matrix into the intermembrane space, creating an electrochemical gradient. Proton return through the F_o portion of Complex V drives conformational changes in F₁ that synthesize ATP. Electron flow, proton pumping, and phosphorylation are therefore functionally coupled.

- Proton-motive force is the coupling intermediate
- F_o conducts protons
- F₁ catalyzes ATP formation
- ETC blockade prevents maintenance of the gradient

Application: Oligomycin inhibition of ATP synthase Tissue vulnerability in severe hypoxia

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 6, oxidative phosphorylation (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Respiratory-chain inhibitors versus uncouplers

A respiratory-chain inhibitor stops electron flow at a defined complex, decreases oxygen consumption, and prevents ATP generation. An uncoupler allows electron transport and oxygen use to continue or accelerate while dissipating the proton gradient as heat, so ATP production falls. Rotenone/amytal inhibit Complex I, antimycin A inhibits Complex III, and cyanide or carbon monoxide inhibit Complex IV.

- Inhibitor: electron flow and oxygen consumption fall
- Uncoupler: oxygen consumption rises while ATP falls
- UCP1 supports physiologic thermogenesis
- DNP is a dangerous synthetic uncoupler

Application: Cyanide poisoning Salicylate toxicity Brown-fat thermogenesis

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 6, inhibitors and uncoupling (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Cytosolic NADH shuttles and reactive oxygen species

The inner mitochondrial membrane lacks a direct NADH transporter, so reducing equivalents enter through the malate-aspartate or glycerol-3-phosphate shuttle. The malate-aspartate shuttle preserves transfer to mitochondrial NADH and the higher ATP yield. Electron leakage can partially reduce oxygen to reactive oxygen species, opposed by superoxide dismutase, catalase, and glutathione peroxidase.

- Shuttles move reducing equivalents, not NADH itself
- Malate-aspartate is more energy efficient
- ROS damage lipids, proteins, and DNA
- Antioxidant enzymes limit mitochondrial oxidant injury

Application: Oxidative stress Mitochondrial disease

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 6, shuttles and ROS (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

09 Carbohydrate structure and stereochemistry

Source basis: Lippincott, Chapter 7; concept-specific teacher references below.

Disaccharides and bonds

Sugar	Building blocks / linkage	Reducing?
Maltose	Glucose- α (14)-glucose	Yes
Cellobiose	Glucose- β (14)-glucose	Yes
Lactose	Galactose- β (14)-glucose	Yes
Sucrose	Glucose- α (12)- β -fructose	No: both anomeric carbons linked
Trehalose	Glucose- α (11)- α -glucose	No: both anomeric carbons linked
Amylopectin / glycogen	α (14) chains, α (16) branches	Glycogen is more highly branched
Cellulose / chitin	β (14) glucose / N-acetylglucosamine	Structural homopolymers

Monosaccharide classification and stereochemistry

Monosaccharides are classified by carbonyl type and carbon number. Epimers differ at one chiral carbon, while enantiomers are mirror images; D-glucose and D-galactose are C4 epimers, and glucose and mannose are C2 epimers. Cyclization creates an anomeric carbon, and alpha and beta forms differ at that carbon.

- Aldose versus ketose
- Epimer differs at one chiral center
- Anomer differs at the anomeric carbon
- D/L does not specify optical rotation

Application: Galactose and fructose metabolic disorders

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 7, monosaccharide structure (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Glycosidic bonds and reducing sugars

A glycosidic bond links the anomeric carbon of a sugar to an oxygen or nitrogen on another molecule. A sugar is reducing when it retains a free anomeric carbon capable of opening to a reactive carbonyl. All monosaccharides are reducing; lactose and maltose are reducing, whereas sucrose is not because both anomeric carbons participate in the bond.

- Free anomeric carbon permits reduction tests
- Glycosidic linkages are named by alpha/beta and carbon positions
- Sucrose is nonreducing
- Benedict reagent detects reducing sugars

Application: Reducing substances in urine Carbohydrate qualification practical

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 7, glycosidic bonds (textbook-confirmed). Carbohydrate Qualification Test Esmaeili Jan 2026 - 21-page teacher practical deck (teacher-confirmed practical context). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Dietary disaccharides and carbohydrate digestion

Starch digestion begins with alpha-amylases, whereas final carbohydrate digestion occurs at the small-intestinal brush border. Lactase, sucrase-isomaltase, and maltase produce glucose, galactose, and fructose for absorption. Undigested carbohydrate retains water and is fermented by colonic bacteria, producing acidic diarrhea, gas, and bloating.

- Lactose = glucose + galactose
- Sucrose = glucose + fructose
- Brush-border enzymes finish digestion
- Lactase is commonly deficient

Application: Lactose intolerance Congenital sucrase-isomaltase deficiency

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 7, digestion of carbohydrates (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Intestinal monosaccharide absorption

Glucose and galactose enter enterocytes apically through sodium-dependent SGLT1, whereas fructose uses facilitated diffusion through GLUT5. All three leave basolaterally mainly through GLUT2. The sodium gradient that powers SGLT1 is maintained indirectly by the basolateral Na⁺/K⁺-ATPase.

- SGLT1 transports glucose and galactose
- GLUT5 transports fructose
- GLUT2 supports basolateral exit
- Na⁺/K⁺-ATPase maintains the driving gradient

Application: Oral rehydration therapy Glucose-galactose malabsorption

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 7, monosaccharide uptake (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Past-paper checkpoints and wording traps

A patient gets abdominal pains and cramps, diarrhea and gas in intestines after consuming milk and dairy products. His problem is related to a defect in which sugar and enzyme below?

Lactose-lactase is the best answer. Monosaccharide composition and glycosidic linkage define carbohydrate behavior.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q37

All the following compounds possess a chiral carbon, except?

Dihydroxyacetone is the best answer. Dihydroxyacetone has no stereogenic carbon.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q43

Which of the following are anomeric isomers?

α-D-Glucose and β-D-glucose is the best answer. Monosaccharide composition and glycosidic linkage define carbohydrate behavior.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q46

Which of the following is a homo-polysaccharide?

Cellulose is the best answer. Cellulose is a homopolymer of glucose.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q56

The enzyme which hydrolyses starch to maltose is

Amylase is the best answer. Amylase cleaves $\alpha(14)$ bonds in starch, producing maltose and oligosaccharides.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q62

Toothpaste and sugar-free gum are probably sweetened with a sugar alcohol made by reducing the aldehyde carbon of xylose to an alcohol. The resulting compound would be named

Xylitol is the best answer. Reduction of xylose yields the sugar alcohol xylitol.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q64

In which of the following disaccharides the glycosidic linkage is in between anomeric carbons of both monosaccharides?

Sucrose is the best answer. Sucrose links both anomeric carbons, making it nonreducing.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q66

The phenomenon of mutarotation is due to which of the following?

Conversion of α -D-Glucose to β -D-Glucose is the best answer. Mutarotation is interconversion of α and β anomers through the open-chain form.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q28

Which of the following is a deoxy sugar?

L-Fucose is the best answer. L-Fucose is 6-deoxy-L-galactose.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q31

Galactonic acid is caused by oxidation of which factor in galactose?

Aldehyde group is the best answer. Monosaccharide composition and glycosidic linkage define carbohydrate behavior. Oxidation of the aldehyde group gives the aldonic acid. An aldehyde contains a carbonyl group, so both the broad and specific choices describe that same site.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q32

The enzyme which hydrolyses Galactose -Glucose linkage is

Lactase is the best answer. Lactase hydrolyzes the $\beta(14)$ galactose-glucose bond of lactose.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q48

Which option is correct for Cellobiose?

It has $\beta(1-4)$ glycosidic bond is the best answer. Cellobiose consists of glucose residues joined by a $\beta(14)$ glycosidic bond.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q50

What units make up Lactose with what bonds?

β -galactose+glucose, $\beta(1-4)$ is the best answer. Lactose is β -galactose linked $\beta(14)$ to glucose.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q52

How many spatial isomers does the linear form of aldohexoses have?

16 is the best answer. An aldohexose has four stereocenters, giving $2^4 = 16$ stereoisomers.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q54

Which of the following produces N-acetyl glucosamine upon hydrolysis?

Chitin is the best answer. Chitin is a polymer of $\beta(14)$ -linked N-acetylglucosamine.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q62

Amylopectin has:

$\alpha(1-4)$ and $\alpha(1-6)$ linkage is the best answer. Amylopectin contains $\alpha(14)$ glucose chains with $\alpha(16)$ branch points.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q61

Which of the following is a pair of non-reducing carbohydrate?

Sucrose and trehalose is the best answer. Sucrose and trehalose have both anomeric carbons tied up in their glycosidic bonds, so neither is reducing.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q46

Which of the following disaccharides contains a beta-glycosidic bond?

Lactose and cellobiose is the best answer. Monosaccharide composition and glycosidic linkage define carbohydrate behavior. The intended comparison is lactose/cellobiose (both $\beta(1,4)$). Sucrose also has a β -D-fructofuranosyl anomeric linkage, so the literal wording also allows lactose/sucrose.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q49

In polysaccharides, monosaccharides are joined by:

Glycosidic bonds is the best answer. Monosaccharides are joined to one another by glycosidic bonds.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q23

The enzyme which hydrolyses Glucose-Fructose linkage is....

Sucrase is the best answer. Sucrase hydrolyzes the glucose-fructose glycosidic bond of sucrose.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q56

Structural polysaccharides include:

Cellulose and chitin is the best answer. Cellulose and chitin are structural polysaccharides; glycogen and starch are storage polysaccharides.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q63

10 Glycoconjugates

Source basis: Lippincott, Chapter 14; concept-specific teacher references below.

Glycosaminoglycan structure and major examples

Glycosaminoglycans are long repeating disaccharide chains whose sulfate and carboxyl groups attract cations and water. Their hydrated gels resist compression and support extracellular matrix. Hyaluronan is unusual because it is unsulfated, is not covalently attached to a core protein, and is synthesized at the plasma membrane.

- Most GAGs are highly negatively charged
- Hydration provides resilience
- Hyaluronan is unsulfated
- Keratan sulfate lacks uronic acid

Application: Cartilage mechanics Heparin anticoagulation

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 14, glycosaminoglycans (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Proteoglycans versus glycoproteins

A proteoglycan contains a core protein bearing long GAG chains, so carbohydrate is usually the dominant mass. Glycoproteins typically contain shorter, branched oligosaccharides and relatively more protein. Proteoglycans chiefly organize hydrated extracellular matrix, while glycoproteins have diverse roles in receptors, adhesion, immunity, and secretion.

- Proteoglycan = core protein plus GAG chains
- Glycoprotein glycans are shorter and branched
- Proteoglycans are carbohydrate-rich
- Both are formed through specific glycosylation pathways

Application: Basement membranes Cell-surface receptors and immune proteins

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 14, proteoglycans and glycoproteins (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

N-linked glycosylation and mannose-6-phosphate targeting

N-linked oligosaccharides attach to asparagine and begin assembly on dolichol in the ER before Golgi processing. Lysosomal hydrolases acquire mannose-6-phosphate in the Golgi, which directs them to lysosomes. Failure to add the tag causes I-cell disease, with hydrolases secreted extracellularly and undegraded substrates accumulating in cells.

- N-linked glycans attach to asparagine
- Dolichol supports ER assembly
- Mannose-6-phosphate targets lysosomal enzymes
- I-cell disease is a targeting defect

Application: I-cell disease Congenital disorders of glycosylation

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 14, glycoprotein synthesis and lysosomal targeting (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Mucopolysaccharidoses: Hurler versus Hunter

Lysosomal defects in GAG degradation cause mucopolysaccharidoses with coarse facial features, organomegaly, skeletal abnormalities, and developmental effects. Hurler syndrome is autosomal recessive alpha-L-iduronidase deficiency and typically has corneal clouding. Hunter syndrome is X-linked iduronate-sulfatase deficiency and generally lacks corneal clouding.

- Both accumulate dermatan and heparan sulfate
- Hurler is autosomal recessive
- Hunter is X-linked
- Corneal clouding favors Hurler

Application: Hurler syndrome Hunter syndrome

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 14, mucopolysaccharidoses (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Past-paper checkpoints and wording traps

Which amino acid is involved in N-glycosidic bond formation?

Asparagine is the best answer. N-linked oligosaccharides attach to the amide nitrogen of asparagine.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q37

All of these polysaccharides are homopolysaccharide except?

Heparin is the best answer. Heparin is a heteropolysaccharide with repeating disaccharide units.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q40

Oligosaccharides linked to proteins are called:

Glycoproteins is the best answer. Proteins bearing covalently attached oligosaccharides are glycoproteins.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q52

Which of the following has more than one type of sugar in its structure?

Hyaluronic acid is the best answer. Hyaluronic acid is a repeating disaccharide of glucuronic acid and N-acetylglucosamine.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q15

11 Dietary lipids and absorption

Source basis: Lippincott, Chapter 15; concept-specific teacher references below.

Lipid classes and fatty-acid structure

Fatty acids contain a carboxyl group and a hydrocarbon chain; saturated fatty acids have no carbon-carbon double bonds, whereas unsaturated fatty acids contain one or more. A cis double bond introduces a kink that reduces packing and generally lowers melting point, while greater chain length and saturation raise melting point. Triacylglycerols and cholesteryl esters are simple storage or transport lipids; phospholipids and glycolipids contain additional polar groups and are major membrane lipids.

- Saturated fatty acids contain no carbon-carbon double bonds
- Cis double bonds disrupt packing and lower melting point
- Longer and more saturated chains generally melt at higher temperatures
- Triacylglycerol is a concentrated hydrophobic energy store
- Complex lipids include additional phosphate, carbohydrate, or nitrogen-containing groups

Application: Membrane fluidity depends partly on fatty-acid chain length and unsaturation Essential omega-6 and omega-3 fatty acids must be obtained from diet

Sources: Lipid structure Esmaeili.pdf, 96-page teacher deck, early fatty-acid and lipid-classification sections. biochemistry-telegram-scope.md: lipid structure, classification, fatty acids, complex lipids, and lipoproteins are confirmed first-semester teacher material.

Enzymatic digestion of dietary lipids

Lingual and gastric lipases begin TAG digestion and are especially useful for short- and medium-chain TAG and in neonates. Pancreatic lipase with colipase performs most adult TAG digestion, hydrolyzing the 1 and 3 positions to yield 2-monoacylglycerol and free fatty acids. Pancreatic esterases and phospholipases digest cholesterol esters and phospholipids.

- Acid lipases start digestion
- Pancreatic lipase is the main adult TAG enzyme
- Colipase anchors lipase at the lipid-water interface
- TAG digestion yields 2-MAG plus free fatty acids

Application: Pancreatic insufficiency Orlistat inhibition of lipases

Sources: Lipid and Lipoprotein Metabolism 1, Fataneh Esmaeili - pp. 4-21 (teacher-confirmed). Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 15 (cross-check). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Bile salts and emulsification

Conjugated bile salts are amphipathic cholesterol derivatives released into the duodenum. Together with intestinal mixing and phospholipids, they emulsify large lipid droplets into smaller particles with greater surface area. Bile salts also remain with digestion products to form mixed micelles for delivery across the unstirred water layer.

- Bile salts are made in liver and stored in gallbladder
- Conjugation increases amphipathic behavior
- Emulsification increases enzyme-accessible area
- Bile flow is required for fat-soluble vitamin absorption

Application: Cholestasis Fat-soluble vitamin deficiency Gallstone disease

Sources: Lipid and Lipoprotein Metabolism 1, Fataneh Esmaeili - pp. 17-24 (teacher-confirmed). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Mixed micelles, enterocyte uptake, and re-esterification

Mixed micelles solubilize long-chain fatty acids, 2-monoacylglycerol, cholesterol, lysophospholipids, and fat-soluble vitamins and carry them to the enterocyte surface. The lipids leave the micelle and enter the cell; the micelle itself is not absorbed. Enterocytes activate long-chain fatty acids and re-esterify complex lipids in smooth ER, whereas short- and medium-chain fatty acids can enter portal blood bound to albumin.

- Micelles cross the unstirred water layer
- Micelles are not endocytosed intact
- Long-chain lipids are re-esterified
- Short/medium chains can use portal blood directly

Application: Fat malabsorption Use of medium-chain TAG in selected malabsorption disorders

Sources: Lipid and Lipoprotein Metabolism 1, Fataneh Esmaeili - pp. 22-27 (teacher-confirmed). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Chylomicron assembly, clearance, and steatorrhea

Enterocytes package re-esterified TAG with apoB-48 through microsomal TAG-transfer protein to form chylomicrons, which enter lymph before blood. In plasma they acquire apoC-II and apoE from HDL; apoC-II activates capillary LPL, and apoE directs remnant uptake by liver. Defective digestion, bile delivery, pancreatic secretion, or mucosal absorption can cause steatorrhea and fat-soluble vitamin deficiency.

- ApoB-48 and MTP are required for assembly
- Chylomicrons travel through lymph
- ApoC-II activates LPL
- ApoE supports remnant uptake

Application: Familial chylomicronemia Cystic fibrosis Steatorrhea

Sources: Lipid and Lipoprotein Metabolism 1, Fataneh Esmaeili - pp. 27-42 (teacher-confirmed). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

12 Fatty acids, triacylglycerols and ketones

Source basis: Lippincott, Chapter 16; concept-specific teacher references below.

Fatty-acid activation and the carnitine shuttle

Acyl-CoA synthetase activates a fatty acid to fatty acyl-CoA at a cost equivalent to two ATP. Long-chain acyl groups enter the mitochondrial matrix through CPT-I, the carnitine-acylcarnitine translocase, and CPT-II. Malonyl-CoA inhibits CPT-I, preventing simultaneous synthesis and oxidation of fatty acids.

- Activation forms fatty acyl-CoA
- CPT-I is the regulated entry step
- Carnitine carries long-chain acyl groups
- Malonyl-CoA inhibits CPT-I

Application: Primary carnitine deficiency CPT deficiency

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 16 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Mitochondrial beta-oxidation and odd-chain fatty acids

Each beta-oxidation cycle performs oxidation, hydration, oxidation, and thiolysis, producing FADH₂, NADH, acetyl-CoA, and a two-carbon-shorter acyl-CoA. Even-chain fatty acids end as acetyl-CoA and cannot produce net glucose. Odd-chain fatty acids also yield propionyl-CoA, which becomes succinyl-CoA through biotin- and B12-dependent steps.

- Occurs mainly in mitochondrial matrix
- Four recurring reactions
- Produces NADH, FADH₂, and acetyl-CoA
- Propionyl-CoA is glucogenic through succinyl-CoA

Application: Methylmalonic acidemia Propionic acidemia

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 16, fatty-acid oxidation (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Ketogenesis and ketone-body utilization

During fasting, hepatic mitochondrial beta-oxidation supplies acetyl-CoA while oxaloacetate is diverted toward gluconeogenesis, favoring ketogenesis. Mitochondrial HMG-CoA synthase is rate limiting for formation of acetoacetate and beta-hydroxybutyrate. Extrahepatic mitochondria use thiophorase to oxidize ketones, but liver lacks thiophorase and RBCs lack mitochondria.

- Liver makes but cannot use ketone bodies
- HMG-CoA synthase is rate limiting
- Brain use rises during prolonged fasting
- Excess ketones cause high-anion-gap acidosis

Application: Diabetic ketoacidosis Starvation adaptation Hypoketotic hypoglycemia

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 16, ketone bodies (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Fatty-acid synthesis and reciprocal control

Cytosolic acetyl-CoA carboxylase uses biotin and ATP to form malonyl-CoA, the committed and rate-limiting step. Fatty-acid synthase uses acetyl and malonyl units plus NADPH to make palmitate. Insulin and citrate activate synthesis, whereas glucagon, epinephrine, AMPK, and long-chain acyl-CoA oppose it; malonyl-CoA simultaneously blocks CPT-I.

- ACC makes malonyl-CoA
- Biotin is required
- FAS mainly produces palmitate
- NADPH supplies reducing power
- Malonyl-CoA coordinates synthesis with oxidation

Application: Hepatic steatosis Metabolic syndrome

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 16, fatty-acid synthesis (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Triacylglycerol synthesis and adipose lipolysis

TAG synthesis esterifies three fatty acyl-CoA molecules to glycerol 3-phosphate; adipocytes depend strongly on glucose-derived DHAP for their glycerol backbone. During fasting, adipose lipases release free fatty acids and glycerol. Fatty acids travel on albumin to oxidative tissues, whereas glycerol goes mainly to liver because adipose tissue has little glycerol kinase.

- Glycerol 3-phosphate supplies the TAG backbone
- Insulin promotes adipose TAG storage
- Fasting activates lipolysis
- Released fatty acids bind albumin
- Glycerol is handled mainly by liver

Application: Fasting hyperlipolysis Insulin resistance

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 16, TAG metabolism (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Fatty-acid oxidation disorders and hypoketotic hypoglycemia

Fatty-acid oxidation supports fasting ATP, hepatic gluconeogenesis, and ketone production. MCAD deficiency impairs medium-chain oxidation and commonly presents after fasting or illness with hypoketotic hypoglycemia, vomiting, lethargy, and dicarboxylic aciduria. Carnitine or shuttle defects similarly limit long-chain oxidation and can affect muscle and heart.

- Low glucose plus low ketones suggests FA-oxidation failure
- MCAD is autosomal recessive
- Avoid fasting
- Dicarboxylic acids reflect omega oxidation

Application: MCAD deficiency Carnitine deficiency Sudden decompensation during illness

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 16, oxidation disorders (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Past-paper checkpoints and wording traps

Which of the following is an essential fatty acid?

Linoleic acid is the best answer. Linoleic and α -linolenic acids are essential because humans cannot introduce the required distal double bonds.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q41

Which of the following fatty acids has the lowest melting point?

Eicosapentaenoic acid is the best answer. More cis double bonds introduce more kinks and lower fatty-acid melting temperature.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q58

Which of the following fatty acid has 16 carbon atoms?

Palmitic acid is the best answer. Palmitic acid is the 16-carbon saturated fatty acid 16:0.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q2

Which of the following statements concerning fatty acids is correct?

One of them is a precursor of prostaglandins is the best answer. Eicosanoids are local signals derived mainly from arachidonic acid.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q18

13 Complex lipids and eicosanoids

Source basis: Lippincott, Chapter 17; concept-specific teacher references below.

Phospholipid structure, membranes, and PIP2 signaling

Phospholipids are amphipathic molecules that form membrane bilayers and also serve as reservoirs for signaling molecules. Receptor-activated phospholipase C cleaves PIP2 to IP3 and DAG. IP3 releases ER Ca²⁺, while DAG activates protein kinase C.

- Polar head and hydrophobic tails
- Phospholipids form bilayers
- PLC cleaves PIP2
- IP3 raises cytosolic Ca²⁺
- DAG activates PKC

Application: Signal-transduction disorders GPI-anchor deficiency in PNH

Sources: Lipid Structure Esmaeili - 96-page teacher deck, phospholipid section (teacher-confirmed). Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 17 (mechanism cross-check). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

DPPC, pulmonary surfactant, and neonatal respiratory distress

Type II pneumocytes secrete surfactant rich in dipalmitoyl phosphatidylcholine, which lowers alveolar surface tension and prevents collapse at end-expiration. Production rises late in gestation. Prematurity can therefore cause neonatal respiratory distress syndrome, treated or prevented with antenatal glucocorticoids and exogenous surfactant as appropriate.

- DPPC is the major surface-active phospholipid
- Made by type II pneumocytes
- Reduces surface tension
- Deficiency causes neonatal RDS

Application: Prematurity Neonatal respiratory distress syndrome

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 17, surfactant (confirmed August 25 scope in tracked bank). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Sphingolipid architecture and sphingolipidoses

Ceramide is the core of sphingomyelin and glycosphingolipids. Lysosomal hydrolase defects cause substrate-specific accumulation: hexosaminidase A deficiency causes Tay-Sachs, sphingomyelinase deficiency causes Niemann-Pick A/B, and glucocerebrosidase deficiency causes Gaucher disease. Clinical pattern recognition should be linked to the accumulated lipid and missing enzyme rather than memorized as isolated names.

- Ceramide is the common sphingolipid core
- Tay-Sachs: GM2 and hexosaminidase A
- Niemann-Pick A/B: sphingomyelinase
- Gaucher: glucocerebrosidase

Application: Cherry-red macula Hepatosplenomegaly Neurodegeneration

Sources: Lipid Structure Esmaeili - sphingolipid section (teacher-confirmed class). Tracked comprehensive bank - Chapter 17, PDF pp. 374-405 (marked confirmed August 25 scope). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Arachidonic acid, cyclooxygenase products, and NSAIDs

Phospholipase A2 releases arachidonic acid from membrane phospholipids. Cyclooxygenases convert it toward prostaglandins, prostacyclin, and thromboxanes; NSAIDs inhibit COX and therefore reduce these mediators. Glucocorticoids act further upstream by inducing lipocortin and reducing phospholipase-A2-mediated release.

- Arachidonate is stored in membrane phospholipids
- COX produces prostanoid precursors
- NSAIDs inhibit COX
- Glucocorticoids reduce arachidonate release

Application: Pain and inflammation Aspirin antiplatelet effect Gastric and renal NSAID toxicity

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 17, eicosanoids (confirmed August 25 scope in tracked bank). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Lipoxygenase, leukotrienes, and aspirin-exacerbated asthma

The 5-lipoxygenase pathway converts arachidonic acid into leukotrienes. Cysteinyl leukotrienes promote bronchoconstriction, mucus production, and vascular permeability. COX inhibition does not block this branch and can divert substrate toward leukotrienes, contributing to aspirin-exacerbated respiratory disease.

- 5-LOX produces leukotrienes
- Leukotrienes are not blocked by NSAIDs
- Cysteinyl leukotrienes constrict bronchi
- Montelukast blocks their receptor

Application: Asthma Aspirin-exacerbated respiratory disease

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 17, leukotrienes (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Past-paper checkpoints and wording traps

An example of glycerophospholipid involved in cell signaling is

Phosphatidylinositol is the best answer. Phosphatidylinositol derivatives generate the IP3 and DAG second-messenger pathway. Phosphatidylinositol derivatives participate in signaling; phosphatidic acid is also a signaling lipid. Both choices are valid under this broad stem.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q55

Dipalmitoyl lecithin acts as

Lung surfactant is the best answer. Dipalmitoyl phosphatidylcholine is the major surface-tension-lowering phospholipid of pulmonary surfactant.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q24

Which of the following belong to glycolipids?

Cerebrosides is the best answer. Cerebrosides are glycosphingolipids consisting of ceramide plus one sugar.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q77

What is the starting compound for the synthesis of sphingolipids?

Ceramide is the best answer. Ceramide is the structural core of sphingolipids. Ceramide is the common branch-point precursor for complex sphingolipids. De novo synthesis begins earlier with serine and palmitoyl-CoA; do not mistake the offered intermediate for the first substrates.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q16

Which of the following phospholipids is the main component of inner mitochondrial membrane?

Cardiolipin is the best answer. Cardiolipin is the signature phospholipid of the inner mitochondrial membrane. Cardiolipin is the characteristic, functionally important inner-mitochondrial phospholipid intended by the exam. It is not necessarily the most abundant phospholipid by mass; “main component” is imprecise.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q29

Which membrane fatty acids serve as precursors for eicosanoids?

Eicosanoids are generated from long-chain polyunsaturated fatty acids released from membrane phospholipids; arachidonic acid is the major C20 precursor. This paper uses eicosanoid loosely for lipid mediators from C20/C22 PUFAs. Strictly, eicosanoids derive from C20 precursors; C22 DHA gives docosanoids. The offered D is the intended broad PUFA choice, not the precise definition.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q36

The backbone of phospholipids is?

L-glycerol 3-phosphate is the best answer. Mammalian glycerophospholipids use the sn-glycerol-3-phosphate backbone. The intended answer describes glycerophospholipids: sn-glycerol-3-phosphate (historically L-glycerol-3-phosphate). Not all phospholipids have glycerol: sphingomyelin has a sphingosine backbone.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q56

Non-steroidal anti-inflammatory drugs (NSAIDs) like aspirin and ibuprofen act by blocking the production of

Prostaglandins is the best answer. Aspirin and ibuprofen inhibit cyclooxygenase and therefore reduce prostaglandin synthesis.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q19

Sphingosine is not a component of....

Cardiolipin is the best answer. Cardiolipin is a glycerophospholipid; ceramide, sphingomyelin, and gangliosides contain a sphingosine-derived backbone.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q13

Which of the following contains an ether-linked alkyl group?

Platelet-activating factor is the best answer. Platelet-activating factor is an ether glycerophospholipid.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q25

14 Cholesterol, lipoproteins and steroids

Source basis: Lippincott, Chapter 18; concept-specific teacher references below.

Lipoprotein traffic

Particle	Main role	Recognition point
Chylomicron	Dietary TAG from intestine to tissues	ApoB-48; apoC-II activates LPL; apoE permits remnant uptake.
VLDL IDL LDL	Hepatic TAG export, then cholesterol delivery	ApoB-100; LDL receptor recognizes apoB-100/apoE.
HDL	Cholesterol transport and exchange	Nascent HDL is formed in liver AND intestine; apoA-I activates LCAT.
Albumin-bound fatty acids	Nonesterified fatty-acid transport	Not a lipoprotein particle; especially relevant during fasting.

Cholesterol synthesis, HMG-CoA reductase, and statins

Cytosolic/ER HMG-CoA reductase converts HMG-CoA to mevalonate using NADPH and is the rate-limiting step of cholesterol synthesis. Insulin and low sterol favor activity, whereas glucagon, AMPK phosphorylation, high sterol, and enzyme degradation reduce it. Statins competitively inhibit the enzyme and increase hepatic LDL-receptor expression, lowering plasma LDL.

- HMG-CoA reductase is rate limiting
- Uses NADPH
- AMPK phosphorylation inhibits it
- Statins increase hepatic LDL clearance

Application: Hypercholesterolemia Statin therapy Atherosclerotic cardiovascular disease

Sources: Lipid Structure Esmaeili - cholesterol/lipoprotein sections (teacher-confirmed class). Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 18 (regulatory cross-check). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Bile-acid synthesis and cholesterol elimination

Humans cannot fully oxidize the steroid nucleus, so cholesterol leaves mainly in bile as cholesterol or bile acids. Cholesterol 7-alpha-hydroxylase is rate limiting for bile-acid synthesis. Conjugation with glycine or taurine makes amphipathic bile salts that emulsify dietary lipid and undergo enterohepatic recycling.

- Biliary secretion is the net elimination route
- 7-alpha-hydroxylase is rate limiting
- Bile salts are amphipathic
- Most bile salts are recycled

Application: Cholestasis Bile-acid sequestrants Gallstones

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 18, bile acids (textbook-confirmed; teacher emphasis uncertain). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Lipoprotein pathways and apolipoprotein map

Chylomicrons carry dietary TAG with apoB-48; VLDL carries hepatic TAG with apoB-100 and becomes IDL then LDL. ApoC-II activates LPL, apoE mediates remnant uptake, apoB-100 binds the LDL receptor, and apoA-I activates LCAT on HDL. HDL also supplies apoC-II and apoE to nascent TAG-rich particles.

- ApoB-48: chylomicron structure
- ApoB-100: VLDL/LDL structure and LDL-receptor ligand
- ApoC-II activates LPL
- ApoE clears remnants
- ApoA-I activates LCAT

Application: Familial chylomicronemia Dysbetalipoproteinemia Familial hypercholesterolemia

Sources: Lipid Structure Esmaeili - 96-page deck, lipoprotein section (teacher-confirmed). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

LDL-receptor uptake and atherosclerotic foam cells

LDL delivers cholesterol through apoB-100 binding to the LDL receptor and receptor-mediated endocytosis. LDL-receptor loss, defective apoB-100 binding, or gain-of-function PCSK9 variants can cause severe familial hypercholesterolemia. Macrophage scavenger receptors take up oxidized LDL without normal cholesterol feedback, creating foam cells and promoting atherosclerotic plaques.

- LDL receptor recognizes apoB-100
- Receptors recycle after endosomal release
- Familial hypercholesterolemia causes early ASCVD
- Scavenger receptors are not normally downregulated by cholesterol

Application: Tendon xanthomas Premature coronary disease PCSK9-targeted therapy

Sources: Lipid Structure Esmaeili - lipoprotein section (teacher-confirmed). Tracked comprehensive bank - Chapter 18 (marked confirmed August 25 scope). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

HDL maturation and reverse cholesterol transport

Lipid-poor apoA-I accepts cellular cholesterol through ABCA1 to form nascent HDL. LCAT esterifies surface cholesterol, allowing cholesteryl ester to enter the HDL core; HDL returns it directly to liver through SR-B1 or transfers it to apoB particles through CETP. ABCA1 deficiency causes Tangier disease with nearly absent HDL.

- ABCA1 loads cholesterol onto apoA-I
- LCAT esterifies HDL cholesterol
- SR-B1 supports selective hepatic uptake
- CETP exchanges HDL cholesteryl ester for TAG

Application: Tangier disease Reverse cholesterol transport

Sources: Lipid Structure Esmaeili - lipoprotein section (teacher-confirmed). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Cholesterol as steroid precursor and acute StAR control

All steroid hormones derive from cholesterol. The acute regulated step is StAR-mediated movement of cholesterol into the inner mitochondrial membrane, where CYP11A1 converts it to pregnenolone. Tissue-specific hydroxylases then generate glucocorticoids, mineralocorticoids, and sex steroids, which act mainly through intracellular nuclear receptors.

- Cholesterol is the common precursor
- StAR controls mitochondrial entry
- CYP11A1 makes pregnenolone
- Steroid receptors regulate transcription

Application: Congenital adrenal hyperplasia Lipoid adrenal hyperplasia Aromatase-inhibitor therapy

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 18, steroid hormones (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Past-paper checkpoints and wording traps

Dietary fats after absorption appear in the circulation as?

Chylomicron is the best answer. Enterocytes package absorbed dietary lipids into chylomicrons.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q42

HDL is synthesized in?

Liver and intestine is the best answer. Nascent HDL is produced by both liver and intestine.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q70

The lipoprotein richest in cholesterol is ...

LDL is the best answer. LDL is the most cholesterol-rich major plasma lipoprotein.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q30

Which of the following lipoproteins is produced via the exogenous lipoprotein pathway?

Chylomicrons is the best answer. Chylomicrons transport dietary lipid through the exogenous pathway.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q39

Which one of the following is assembled and secreted by the liver?

Very Low-Density Lipoprotein (VLDL) is the best answer. The liver packages endogenous triacylglycerol into VLDL for secretion. The liver secretes VLDL and also nascent HDL. VLDL is the intended triglyceride-export answer, but the unrestricted stem permits both HDL and VLDL.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q55

15 Insulin and glucagon

Source basis: Lippincott, Chapter 23; concept-specific teacher references below.

Insulin biosynthesis and glucose-stimulated secretion

Preproinsulin becomes proinsulin and then insulin plus C-peptide in secretory granules. Beta-cell glucose metabolism raises ATP, closes ATP-sensitive K⁺ channels, depolarizes the membrane, and opens voltage-gated Ca²⁺ channels to trigger exocytosis. Endogenous insulin secretion releases equimolar C-peptide, unlike injected insulin.

- Preproinsulin to proinsulin to insulin plus C-peptide
- ATP closes KATP channels
- Ca²⁺ triggers granule release
- C-peptide marks endogenous secretion

Application: Insulinoma Sulfonylurea action Distinguishing injected insulin

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 23 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Insulin receptor signaling and metabolic effects

The insulin receptor is a receptor tyrosine kinase that signals through IRS proteins, PI3K, and Akt. Insulin recruits GLUT4 to skeletal-muscle and adipose membranes and promotes glycogen, lipid, and protein synthesis while suppressing gluconeogenesis and lipolysis. Liver uses GLUT2 rather than insulin-recruited GLUT4, but insulin still changes hepatic enzyme activity and gene expression.

- Insulin receptor is a receptor tyrosine kinase
- GLUT4 is in muscle and adipose tissue
- Insulin promotes storage and synthesis
- Insulin suppresses hepatic glucose output

Application: Insulin resistance Type 2 diabetes

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 23 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Glucagon signaling and counterregulatory hormones

Glucagon binds a hepatic Gs-coupled receptor, raises cAMP, and activates PKA to promote glycogenolysis and gluconeogenesis while suppressing glycolysis. Skeletal muscle lacks a meaningful glucagon response and mobilizes glycogen mainly through epinephrine and contraction signals. Glucagon and epinephrine provide rapid defense against falling glucose, with cortisol and growth hormone acting more slowly.

- Glucagon uses Gs-cAMP-PKA
- Primary target is liver
- No direct glucagon-driven muscle glycogenolysis
- Counterregulation protects against hypoglycemia

Application: Glucagon rescue for severe hypoglycemia Fasting response

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 23 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Hypoglycemia recognition, mechanism, and biochemical diagnosis

Falling plasma glucose first triggers counterregulatory hormones and autonomic symptoms such as tremor, palpitations, anxiety, and sweating; more severe or prolonged deficiency deprives the brain of fuel and causes confusion, abnormal behavior, seizures, or coma. Ethanol can cause fasting hypoglycemia because its high NADH state diverts pyruvate to lactate and oxaloacetate to malate, limiting gluconeogenesis. During hypoglycemia, high insulin with high C-peptide supports endogenous secretion, whereas injected insulin raises insulin but suppresses C-peptide.

- Autonomic symptoms are early warning signs
- Neuroglycopenia causes confusion, seizures, and coma
- Ethanol inhibits gluconeogenesis through a high-NADH redox shift
- C-peptide distinguishes endogenous from injected insulin
- Interpret insulin and C-peptide only in the context of documented low glucose

Application: Insulinoma versus exogenous-insulin exposure Alcohol-associated fasting hypoglycemia Urgent recognition of neuroglycopenia

Sources: exam-scope.md (updated 2026-08-20): the active August 25 scope explicitly includes insulin, glucagon, and hypoglycemia. Lippincott's Illustrated Reviews: Biochemistry, 6e, Chapter 23, hypoglycemia mechanisms and evaluation.

16 Fed-fast integration

Source basis: Lippincott, Chapter 24; concept-specific teacher references below.

Absorptive-state metabolism by organ

After a carbohydrate-containing meal, high insulin and low glucagon promote hepatic glycolysis, glycogen synthesis, fatty-acid synthesis, and TAG export. Adipose tissue increases GLUT4-mediated glucose uptake and TAG storage, while muscle takes up glucose and amino acids for glycogen and protein synthesis. Brain primarily uses glucose, and RBCs always rely on glycolysis.

- High insulin-to-glucagon ratio
- Liver stores glucose and makes lipid
- Adipose stores TAG
- Muscle makes glycogen and protein
- RBCs remain glycolytic

Application: Postprandial hyperglycemia Insulin action

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 24 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Early fasting: glycogenolysis and gluconeogenesis

In early fasting, lower insulin and higher glucagon cause hepatic glycogenolysis to maintain blood glucose. As hepatic glycogen wanes, gluconeogenesis from lactate, glycerol, and glucogenic amino acids becomes dominant. Adipose lipolysis supplies fatty acids for tissue and hepatic energy plus glycerol for glucose synthesis.

- Glycogenolysis dominates first
- Gluconeogenesis rises with fasting duration
- Lipolysis supplies fatty acids and glycerol
- Liver preserves glucose for dependent tissues

Application: Fasting hypoglycemia Alcohol-associated impaired gluconeogenesis

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 24 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Prolonged fasting and protein-sparing ketone adaptation

With prolonged fasting, hepatic ketone production rises and the brain increasingly oxidizes ketones. This reduces the brain's glucose demand and therefore decreases the amount of amino acid that must be sacrificed for gluconeogenesis. RBCs still require glucose and produce lactate because they lack mitochondria.

- Brain progressively uses ketones
- Ketones spare muscle protein
- RBCs cannot use ketones
- Liver makes but does not consume ketones

Application: Starvation Ketogenic adaptation

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 24 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Tissue-specific fuel constraints

RBCs lack mitochondria and need glucose for glycolysis. Liver can oxidize fatty acids and make ketones but cannot use ketones because it lacks thiophorase; muscle uses fatty acids and ketones according to state. Brain normally relies on glucose but uses substantial ketones during prolonged fasting, while long-chain fatty acids do not cross the blood-brain barrier efficiently.

- RBC: glucose only for ATP
- Liver: produces ketones
- Brain: glucose, then more ketones in fasting
- Muscle: flexible fuel use

Application: Hypoglycemic neurologic symptoms Ketoacidosis

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 24 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

17 Diabetes mellitus

Source basis: Lippincott, Chapter 25; concept-specific teacher references below.

Type 1 versus type 2 diabetes pathophysiology

Type 1 diabetes results from autoimmune beta-cell destruction and absolute insulin deficiency. Early type 2 diabetes combines insulin resistance with compensatory hyperinsulinemia, followed by progressive beta-cell failure. Both increase hepatic glucose output and impair peripheral glucose handling, but severe unopposed lipolysis and ketogenesis are characteristic of untreated type 1 disease.

- Type 1: absolute insulin deficiency
- Type 2: insulin resistance plus beta-cell dysfunction
- Both cause hyperglycemia
- Type 1 has greater DKA risk

Application: Autoimmune type 1 diabetes Metabolic syndrome and type 2 diabetes

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 25 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Diagnosis and glycemic monitoring

Diabetes can be diagnosed with reproducibly elevated fasting plasma glucose, HbA1c, oral-glucose-tolerance values, or classic symptoms with a high random glucose. HbA1c reflects nonenzymatic glycation over the approximate lifespan of circulating RBCs and therefore represents roughly two to three months of glycemia. Altered RBC survival can make HbA1c misleading.

- Repeat abnormal testing when symptoms are absent
- HbA1c reflects recent months
- RBC lifespan affects HbA1c
- Diagnosis and treatment targets are not identical

Application: Diabetes screening Hemoglobinopathy or hemolysis affecting HbA1c

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 25 (textbook/tracked-bank; current thresholds require guideline verification). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

DKA versus hyperosmolar hyperglycemic state

Severe insulin deficiency activates adipose lipolysis and hepatic ketogenesis, causing DKA with high-anion-gap metabolic acidosis. HHS typically retains enough insulin to restrain major ketogenesis but not enough to prevent profound hyperglycemia and osmotic diuresis. Both cause dehydration and require urgent fluid, electrolyte, and insulin management.

- DKA has significant ketonemia and acidosis
- HHS has more extreme hyperosmolality
- Osmotic diuresis drives dehydration
- Potassium balance requires close monitoring

Application: Diabetic ketoacidosis Hyperosmolar hyperglycemic state

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 25 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Biochemical basis of chronic diabetic complications

Chronic hyperglycemia forms advanced glycation end products that cross-link matrix proteins, trap plasma proteins, and activate inflammatory receptors. Sorbitol accumulation and oxidative stress contribute in tissues with insulin-independent glucose uptake. These mechanisms participate in retinopathy, nephropathy, neuropathy, and accelerated atherosclerosis.

- AGEs alter proteins and vessels
- Sorbitol creates osmotic stress
- Microvascular disease affects retina, kidney, and nerves
- Cardiovascular risk is increased

Application: Diabetic retinopathy Nephropathy Neuropathy Atherosclerotic disease

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 25 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

18 Obesity and energy balance

Source basis: Lippincott, Chapter 26; concept-specific teacher references below.

Obesity measurement and metabolic risk

BMI is weight in kilograms divided by height in meters squared and is a screening measure rather than a direct measure of adiposity. Visceral/upper-body fat is more strongly associated with insulin resistance, dyslipidemia, fatty liver, and cardiovascular disease than gluteofemoral fat. Waist measures therefore add risk information beyond BMI.

- BMI = kg/m^2
- BMI 30 or greater conventionally indicates obesity
- Visceral fat carries higher metabolic risk
- Interpret BMI in clinical context

Application: Metabolic syndrome NAFLD Cardiovascular risk

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 26 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Leptin, ghrelin, and hypothalamic appetite circuits

Leptin from adipose tissue signals energy stores and normally suppresses orexigenic NPY/AgRP neurons while favoring anorexigenic pathways. Ghrelin rises before meals and stimulates appetite. Common obesity usually features high leptin with central leptin resistance, while weight loss lowers leptin and increases hunger signals.

- Leptin reflects adipose stores
- Common obesity has leptin resistance
- Ghrelin stimulates appetite
- NPY/AgRP neurons are orexigenic

Application: Adaptive hunger after weight loss Rare congenital leptin deficiency

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 26 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Adipose inflammation, adiponectin, and insulin resistance

Expanded visceral adipose tissue releases excess fatty acids and proinflammatory cytokines and recruits immune cells. Adiponectin, which normally supports insulin sensitivity and fatty-acid oxidation, generally falls in obesity. These changes promote hepatic and muscle insulin resistance and contribute to type 2 diabetes.

- Adipose tissue secretes adipokines
- Adiponectin falls in obesity
- Inflammation promotes insulin resistance
- Free-fatty-acid flux worsens ectopic lipid storage

Application: Type 2 diabetes NAFLD Metabolic syndrome

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 26 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

19 Nutrition

Source basis: Lippincott, Chapter 27; concept-specific teacher references below.

EAR, RDA, AI, and tolerable upper intake level

The EAR estimates the intake meeting the needs of half a defined healthy group, while the RDA is set higher to cover nearly all. AI is used when evidence is insufficient to establish an EAR and RDA. The tolerable upper intake level estimates the highest chronic intake unlikely to create adverse effects and is not a recommended target.

- EAR covers about half of a group
- RDA covers nearly all healthy individuals
- AI substitutes when evidence is insufficient
- UL addresses toxicity risk

Application: Dietary counseling Supplement toxicity

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 27 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Energy density and total energy expenditure

Carbohydrate and protein provide about 4 kcal/g, fat about 9 kcal/g, and ethanol about 7 kcal/g. Resting/basal expenditure is usually the largest component of daily energy use in sedentary adults. Physical activity is the most variable component, and protein has the greatest thermic effect of feeding.

- Fat: 9 kcal/g
- Alcohol: 7 kcal/g
- Carbohydrate/protein: 4 kcal/g
- Basal metabolism is usually the largest component
- Protein has high thermic effect

Application: Weight-management counseling Nutrition support calculations

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 27 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Essential fatty acids, limiting amino acids, and nitrogen balance

Linoleic acid and alpha-linolenic acid are the parent essential fatty acids because humans cannot introduce double bonds beyond carbon 9. A limiting amino acid is the essential amino acid present in the lowest amount relative to need. Positive nitrogen balance occurs in growth, pregnancy, or recovery, whereas illness, trauma, or inadequate protein can produce negative balance.

- Linoleic acid is omega-6
- Alpha-linolenic acid is omega-3
- Essential amino acids must come from diet
- Nitrogen balance reflects protein gain or loss

Application: Growth and pregnancy Catabolic illness Essential-fatty-acid deficiency

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 27 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Kwashiorkor versus marasmus

Kwashiorkor features disproportionate protein deficiency with edema, fatty liver, skin/hair changes, and low albumin. Marasmus reflects severe total energy deficiency with marked wasting of muscle and subcutaneous fat but usually no prominent edema. Mixed phenotypes occur, so real patients require full nutritional and medical assessment.

- Kwashiorkor: edema and fatty liver
- Marasmus: severe wasting
- Low albumin contributes to edema
- Infection commonly complicates both

Application: Childhood malnutrition Refeeding risk

Sources: Lippincott's Illustrated Reviews: Biochemistry, 6e - Chapter 27 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

20 Vitamins

Source basis: Lippincott, Chapter 28; concept-specific teacher references below.

Vitamins: function first, then deficiency

Vitamin / active form	Central role	Exam association
B1 / TPP	Oxidative decarboxylation; transketolase	Beriberi, Wernicke-Korsakoff; PDH, α -KGDH and branched-chain α -ketoacid DH.
B2 / FMN, FAD	Redox electron transfer	Cheilosis, glossitis; light-sensitive riboflavin can be depleted during phototherapy.
B3 / NAD ⁺ , NADP ⁺	Redox and other NAD-dependent reactions	Pellagra: dermatitis, diarrhea, dementia. Tryptophan-to-niacin synthesis requires B6.
B5 / coenzyme A, ACP	Acyl-group transfer	CoA contains pantothenate; not TPP or PLP.
B6 / PLP	Transamination, decarboxylation and other amino-acid reactions	Sideroblastic anemia, neuropathy/seizures; excess B6 can also cause neuropathy.
B7 / biotin	Carboxylation	Pyruvate, acetyl-CoA, propionyl-CoA and methylcrotonyl-CoA carboxylases; avidin binds biotin.
B9 / THF derivatives	One-carbon transfer; DNA synthesis	Megaloblastic anemia; homocysteine increases, methylmalonate does not.
B12 / methyl- and adenosylcobalamin	Methionine synthase; methylmalonyl-CoA mutase	Megaloblastic anemia plus possible neurologic disease; homocysteine and methylmalonate increase.
C / ascorbate	Collagen hydroxylation, antioxidant, iron absorption	Scurvy; deficient hydroxylation weakens collagen.
A / retinal, retinoic acid	Vision and gene regulation	Night blindness / xerophthalmia; excess preformed vitamin A is toxic and teratogenic.
D / calcitriol	Calcium/phosphate regulation	25-OH-D is the major circulating status marker; 1,25-(OH) ₂ -D is the active hormone.
E / tocopherols	Lipid-phase antioxidant	Hemolysis and neurologic dysfunction in deficiency.
K / reduced vitamin K	γ -carboxylation of glutamate residues	Clotting factors II, VII, IX, X and proteins C/S; warfarin blocks vitamin-K recycling.

Vitamin solubility, coenzymes, storage, and toxicity

Most B vitamins act as coenzyme precursors, whereas vitamins A and D also have hormone-like transcriptional roles. Water-soluble vitamins generally have limited storage and faster deficiency, with B12 as a major exception. Fat-soluble vitamins require normal lipid absorption, are stored more readily, and therefore have greater potential for accumulation and toxicity.

- B and C are water soluble
- A, D, E, and K are fat soluble
- Coenzymes may be loosely or tightly bound
- Fat malabsorption threatens A/D/E/K
- Storage increases toxicity potential

Application: Cholestatic vitamin deficiency Supplement toxicity

Sources: Vitamin 2024, TUMS Clinical Biochemistry - pp. 4-12 (teacher-confirmed). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Thiamine (B1): TPP reactions and neurologic deficiency

Thiamine pyrophosphate supports oxidative decarboxylation by PDH, alpha-ketoglutarate dehydrogenase, and branched-chain alpha-ketoacid dehydrogenase, plus transketolase reactions. Deficiency impairs glucose oxidation and can raise lactate. Wernicke encephalopathy presents with confusion, ataxia, and ocular abnormalities and requires thiamine before glucose in an at-risk patient.

- Active form is TPP
- Required by three major dehydrogenase complexes
- Required by transketolase
- Deficiency causes Wernicke-Korsakoff and beriberi

Application: Alcohol-use disorder Refeeding Lactic acidosis

Sources: Vitamin 2024 - water-soluble vitamin section, pp. 13-45 (teacher-confirmed). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Riboflavin (B2) and niacin (B3) in redox metabolism

Riboflavin forms FMN and FAD, tightly associated redox cofactors in dehydrogenases and the respiratory chain. Niacin forms NAD⁺ and NADP⁺; NADH is used mainly for ATP-generating oxidation, while NADPH supports reductive synthesis and antioxidant defense. Niacin deficiency causes pellagra with dermatitis, diarrhea, and dementia.

- B2 gives FMN and FAD
- B3 gives NAD and NADP
- NADH favors energy production
- NADPH favors reductive work
- Pellagra is niacin deficiency

Application: Pellagra Hartnup disease Carcinoid-associated niacin depletion

Sources: Vitamin 2024 - water-soluble vitamin section, pp. 13-45 (teacher-confirmed). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Pantothenate, pyridoxine, and biotin reaction patterns

Pantothenate is part of CoA and acyl-carrier protein, supporting acyl-group transfer. Pyridoxal phosphate from B6 supports amino-acid transamination and decarboxylation and glycogen phosphorylase; isoniazid can cause functional B6 deficiency. Biotin carries CO₂ in carboxylase reactions, and avidin in raw egg white can impair its absorption.

- B5 supports acyl transfer
- PLP supports amino-acid reactions
- Isoniazid can cause B6 deficiency
- Biotin supports carboxylation
- Avidin binds biotin

Application: Isoniazid neuropathy prevention Multiple carboxylase deficiency

Sources: Vitamin 2024 - water-soluble vitamin section, pp. 13-45 (teacher-confirmed). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Folate and B12 one-carbon metabolism

Tetrahydrofolate transfers one-carbon units for nucleotide synthesis, while B12 is needed for methionine synthase and methylmalonyl-CoA mutase. Both deficiencies cause megaloblastic anemia and elevated homocysteine, but only B12 deficiency raises methylmalonic acid and causes characteristic neurologic injury. B12 absorption requires gastric intrinsic factor and terminal ileum.

- Folate supports nucleotide synthesis
- B12 requires intrinsic factor
- Both can cause megaloblastic anemia
- Neurologic deficits and methylmalonate favor B12 deficiency
- Folate prevents neural-tube defects

Application: Pernicious anemia Neural-tube defects Methotrexate therapy

Sources: Vitamin 2024 - folate p. 34; B12 pp. 37-39 (teacher-confirmed). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Vitamins A and D: vision, transcription, and calcium balance

Retinal participates in the visual cycle and retinoic acid regulates transcription through nuclear receptors; vitamin-A deficiency causes night blindness and may progress to xerophthalmia. Vitamin D3 is made in skin, 25-hydroxylated in liver, and 1-alpha-hydroxylated in kidney to calcitriol. Calcitriol supports calcium/phosphate balance and bone mineralization; deficiency causes rickets/osteomalacia and excess causes hypercalcemia.

- 11-cis-retinal supports vision
- Retinoic acid controls transcription
- Vitamin D activation uses skin, liver, kidney
- Calcitriol increases calcium/phosphate availability
- A and D can be toxic in excess

Application: Night blindness Rickets and osteomalacia Vitamin-D toxicity

Sources: Vitamin 2024 - vitamin A pp. 46-55; vitamin D pp. 56-63 (teacher-confirmed). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Vitamins E, K, and C: antioxidant, coagulation, and collagen roles

Vitamin E is a lipid-phase antioxidant that protects membranes and RBCs from peroxidation; deficiency is most likely with fat malabsorption and can cause hemolysis or neurologic dysfunction. Vitamin K supports gamma-carboxylation of glutamate residues in clotting and bone proteins, and deficiency prolongs PT early because factor VII has a short half-life. Vitamin C supports collagen hydroxylation and iron absorption; deficiency causes scurvy with poor wound healing and bleeding.

- Vitamin E protects membranes
- Vitamin K enables gamma-carboxylation
- PT rises early in vitamin-K deficiency
- Vitamin C supports collagen hydroxylation
- Fat malabsorption threatens E and K

Application: Hemolysis in vitamin-E deficiency Warfarin mechanism Scurvy

Sources: Vitamin 2024 - vitamin C pp. 40-45; vitamins E/K pp. 63-70 (teacher-confirmed). exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Past-paper checkpoints and wording traps

Which of the following vitamins provides the coenzyme for γ -carboxylation reactions?

Menaquinone is the best answer. Fat-soluble vitamins A, D, E, and K have distinct stored functions and deficiency patterns.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q7

How many vitamins are classified as fat-soluble?

4 is the best answer. There are four fat-soluble vitamins: A, D, E, and K.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q38

Which of the following is correctly matched with the vitamin it is derived from?

Pyridoxal phosphate-Vitamin B6 is the best answer. Pyridoxal phosphate is the active coenzyme form of vitamin B6.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q51

Which vitamin is a methyl group acceptor?

Vitamin B12 is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q52

Which of the following is not a clinical outcome of niacin deficiency?

Night blindness is the best answer. Niacin deficiency causes pellagra-dermatitis, diarrhea, and dementia-not night blindness.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q54

Which of these is a vitamin A precursor?

Beta-carotene is the best answer. β -Carotene can be cleaved to form vitamin A.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q59

Which of the following is the active form of pantothenic acid?

Coenzyme A is the best answer. Pantothenate is incorporated into coenzyme A.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q69

Compounds that are needed for enzymes to function properly are?

Vitamins is the best answer. Many vitamins are precursors of coenzymes required for enzyme function.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q75

A 60-year old female presents with severe back pain for the past week, she is diagnosed with osteoporosis, a condition resulting from calcium depletion in bones. As treatment which of the following vitamins can be prescribed?

Vit D is the best answer. Vitamin D supports calcium and phosphate absorption and normal bone mineralization.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q76

Which vitamin(s) play important roles in the metabolism of branched-chain amino acids and odd-carbon fatty acids?

Biotin supports propionyl-CoA carboxylase, shared by odd-chain fatty-acid and several amino-acid catabolic routes. B6 and B1 participate elsewhere in branched-chain amino-acid breakdown; they do not supply this shared carboxylase. B12 is also important downstream but is not offered.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q81

Which vitamin deficiency is seen in infants with jaundice who undergo phototherapy?

B2 is the best answer. Phototherapy can degrade riboflavin, making vitamin B2 deficiency a recognized risk.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q1

Which of the following vitamins is involved in the structure of coenzyme A?

Pantothenate (B5) is the best answer. Pantothenate, vitamin B5, is a structural component of coenzyme A.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q6

In vitamin B6 deficiency, which other vitamin deficiency may also be observed?

B3 is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q10

In which reaction does biotin participate?

Carboxylation is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q17

Which on of the following is the main form of vitamin D in plasma?

25-Hydroxycholecalciferol (25(OH)-D3) is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q18

Which of the following vitamins is directly involved in the conversion of homocysteine to methionine?

B12 (Cobalamin) is the best answer. Vitamin B12 is the cofactor for methionine synthase.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q24

Which vitamin deficiency cause hemolytic anemia?

E is the best answer. Vitamin E deficiency permits oxidative membrane damage and can cause hemolytic anemia.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q25

Which vitamin has thiazole and pyrimidine ring in its structure?

B1 (Thiamine) is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q45

Which vitamin is involved in the synthesis of collagen protein?

C is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q53

Deficiency of which one of these vitamins is the major cause of blindness?

Vitamin A is the best answer. Vitamin A is required for retinal function and epithelial integrity; deficiency can progress from night blindness to xerophthalmia.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q41

Which vitamin is a major lipid-soluble antioxidant in cell membrane?

Vitamin E is the best answer. Vitamin E is the principal lipid-phase antioxidant protecting membranes from peroxidation.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q17

All of the following vitamins provide the cofactor for pyruvate dehydrogenase, except...

Folate is the best answer. Pyruvate dehydrogenase requires TPP, lipoate, CoA, FAD, and NAD⁺; folate is not one of its cofactors.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q45

Which vitamin requires cobalt for its activity?

Vitamin B12 is the best answer. Cobalamin is vitamin B12; cobalt is the central metal in its corrin ring.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q11

Which is incorrect statement about vitamin C?

It can be synthesized in the body is the best answer. Humans lack gulonolactone oxidase and therefore cannot synthesize vitamin C.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q38

Deficiency of which one of these vitamins may lead to megaloblastic anemia?

Vitamin B12 is the best answer. Vitamin B12 or folate deficiency impairs DNA synthesis and causes megaloblastic anemia.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q57

In scurvy, which amino acid that is normally a part of collagen is not synthesized?

Hydroxyproline is the best answer. Vitamin C deficiency impairs hydroxylation of proline and lysine in collagen.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q48

Which of the following vitamins provides the cofactor for transamination of amino acids?

Pyridoxine is the best answer. Pyridoxal phosphate, derived from vitamin B6, is the cofactor for aminotransferases.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q6

Thiamine deficiency leads to _____?

Beri-beri is the best answer. Thiamine deficiency causes beriberi and may cause Wernicke-Korsakoff syndrome.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q2

21 DNA structure, replication and repair

Source basis: Lippincott, Chapter 29; concept-specific teacher references below.

DNA repair: do not mix organisms

Mechanism	What is repaired	Key distinction
Base-excision repair	Small base lesions, e.g. uracil in DNA	DNA glycosylase removes the base first; AP processing, resynthesis, ligation follow.
Nucleotide-excision repair	Bulky helix-distorting lesions	Human XP commonly involves XPA-XPG; bacterial UvrABC is not a human gene. XP variant: POLH.
Mismatch repair	Replication mismatches missed by proofreading	Recognizes the newly synthesized strand; human deficiency underlies Lynch syndrome.
Polymerase proofreading	Wrong newly incorporated nucleotide	3'5' exonuclease; DNA synthesis itself is 5'3'.
Homologous recombination / NHEJ	Double-strand breaks	HR uses homologous template; NHEJ directly rejoins ends and may introduce sequence changes.

Nucleosides, nucleotides, and the phosphodiester backbone

A nucleoside is a nitrogenous base joined to a pentose sugar, whereas a nucleotide adds one or more phosphates. Purines are adenine and guanine; pyrimidines are cytosine, thymine, and uracil. Within one nucleic-acid strand, the 3' hydroxyl of one sugar is joined to the 5' phosphate of the next nucleotide by a phosphodiester bond, giving the strand directionality.

- Nucleoside = base + sugar; nucleotide = nucleoside + phosphate.
- Ribose has a 2'-OH, whereas deoxyribose has 2'-H.
- A and G are purines; C, T, and U are pyrimidines.
- The backbone linkage is 3'5' phosphodiester, and sequences are written 5'3'.

Application: Nucleoside and nucleotide analogs are used as antiviral and anticancer drugs because they can inhibit polymerases or terminate chains.

Sources: Teacher: Deoxyribonucleic Acid(DNA)Structure (2).pdf, pp. 6-14 and 25-28. Lippincott Illustrated Reviews: Biochemistry, 6e, Chapter 29, DNA primary structure and polarity.

Complementary antiparallel DNA, helix forms, and melting

Double-stranded DNA contains complementary strands running in opposite 5'3' directions. Adenine pairs with thymine through two hydrogen bonds and guanine pairs with cytosine through three, so Chargaff's relation is $A = T$ and $G = C$. B-DNA is the usual right-handed physiologic form; heat, extreme pH, or low ionic strength denatures DNA, and greater GC content generally raises its melting temperature.

- The strands are complementary and antiparallel.
- A-T has two hydrogen bonds; G-C has three.
- Base stacking also makes a major contribution to duplex stability.
- B-DNA is physiologic and right-handed; Z-DNA is left-handed.

- Denaturation is strand separation without backbone hydrolysis and can be reversed by controlled annealing.

Application: Hybridization-based diagnostics and PCR primer design depend on complementarity and melting temperature.

Sources: Teacher: Deoxyribonucleic Acid(DNA)Structure (2).pdf, pp. 22-34. Lippincott 6e, Chapter 29, DNA secondary structure, denaturation, and A/B/Z forms.

Nucleosomes, histone H1, and chromatin accessibility

A nucleosome core contains two copies each of H2A, H2B, H3, and H4 around which DNA is wrapped. Histone H1 binds linker DNA and promotes higher-order compaction but is not part of the core octamer. Acetylation of lysine residues reduces histone positive charge, loosens DNA association, and generally makes chromatin more transcriptionally accessible.

- Core octamer = $2 \times$ H2A, H2B, H3, and H4.
- Histones bind negatively charged DNA because they are rich in lysine and arginine.
- H1 is the linker histone.
- Open euchromatin is generally more transcriptionally active than compact heterochromatin.
- Histone-tail modifications can recruit proteins that activate or repress genes.

Application: Histone deacetylase inhibitors are used in selected malignancies because chromatin state can alter gene expression.

Sources: Teacher: Deoxyribonucleic Acid(DNA)Structure (2).pdf, pp. 40-43. Lippincott 6e, Chapter 29, chromatin packaging and nucleosome organization.

Semiconservative, bidirectional replication from origins

DNA replication is semiconservative because each daughter duplex retains one parental strand and contains one newly synthesized strand. Replication starts at defined origins and usually proceeds bidirectionally through replication forks. Eukaryotic chromosomes use many licensed origins so their large linear genomes can be copied completely during S phase.

- Each parental strand is a template for a complementary daughter strand.
- An origin opens into a replication bubble with two forks.
- Bacterial chromosomes usually use one main origin; eukaryotic chromosomes use many.
- Origin firing is restricted so each segment is copied once per cell cycle.

Application: Replication stress and defective origin control contribute to genomic instability in cancer.

Sources: Teacher: Deoxyribonucleic Acid(DNA Replication (1)).pptx, slides 3-11 and 53-54. Lippincott 6e, Chapter 29, semiconservative and bidirectional replication.

Helicase, single-strand binding proteins, and topoisomerases

Helicase separates parental strands, while single-strand DNA-binding proteins keep the templates apart and protect them. Unwinding produces positive supercoiling ahead of the fork, which topoisomerases relieve by transiently breaking and resealing DNA. Type I topoisomerases cut one strand without ATP, whereas type II enzymes cut both strands and require ATP.

- DnaB is the principal bacterial replicative helicase.
- SSB prevents reannealing and protects exposed single-stranded DNA.
- Topoisomerase I changes linking number in steps of one.
- Topoisomerase II changes linking number in steps of two and includes bacterial DNA gyrase.

Application: Fluoroquinolones inhibit bacterial gyrase/topoisomerase IV; several anticancer drugs target human topoisomerases.

Sources: Teacher: Deoxyribonucleic Acid(DNA Replication (1).pptx, slides 15 onward in the fork-machinery block. Lippincott 6e, Chapter 29, prepriming complex and topoisomerases.

5'3' synthesis, primers, and leading versus lagging strands

DNA polymerases add nucleotides only to a free 3'-OH, so every new strand grows 5'3' and requires a primer. The leading strand is synthesized continuously toward the fork, whereas the lagging strand is synthesized away from the fork as Okazaki fragments. Primase makes the RNA primers, and the repeated primers on the lagging strand must later be removed and the fragments joined.

- Polymerase reads template 3'5' while synthesizing 5'3'.
- A primer supplies the initial paired 3'-OH.
- Leading synthesis is continuous; lagging synthesis is discontinuous.
- Okazaki fragments arise from antiparallel template geometry.

Sources: Teacher: Deoxyribonucleic Acid(DNA Replication (1).pptx, strand-synthesis and Okazaki-fragment slides. Lippincott 6e, Chapter 29, DNA synthesis directionality and primer requirement.

Polymerase specialization, proofreading, primer removal, and ligation

Replicative polymerases extend DNA processively, and their 3'5' exonuclease activity removes a newly misincorporated nucleotide for proofreading. In bacteria, DNA polymerase I uses 5'3' exonuclease activity to remove RNA primers and fills the resulting gaps with DNA. DNA ligase then seals the remaining nick by creating the final phosphodiester bond.

- Bacterial DNA polymerase III performs most fork elongation.
- Proofreading exonuclease is 3'5'.
- Bacterial DNA polymerase I removes primers with 5'3' exonuclease.
- DNA ligase seals, but does not fill, a nick.
- PCNA is the eukaryotic sliding clamp.

Application: Polymerase proofreading defects produce mutator phenotypes and increase cancer susceptibility.

Sources: Teacher: Deoxyribonucleic Acid(DNA Replication (1).pptx, polymerase, exonuclease, and ligase block. Lippincott 6e, Chapter 29, polymerases I/III, proofreading, and ligation.

Eukaryotic polymerases and the telomere end-replication problem

In eukaryotes, polymerase α works with primase to initiate synthesis, while high-processivity polymerases extend the leading and lagging strands with PCNA as their clamp. Linear chromosome ends cannot be completely copied by ordinary lagging-strand synthesis, causing telomere shortening in most somatic cells. Telomerase solves this problem using an intrinsic RNA template and a reverse-transcriptase protein subunit to extend the 3' chromosome end.

- Polymerase α /primase initiates; polymerases ϵ and δ perform most nuclear elongation.
- Polymerase γ replicates mitochondrial DNA.
- Telomerase contains RNA plus telomerase reverse transcriptase.
- Telomerase is active in germ/stem cells and commonly reactivated in cancer.
- Progressive telomere shortening contributes to replicative senescence.

Application: Telomerase reactivation supports unlimited proliferation in many cancers. Telomere-maintenance defects cause bone-marrow failure and other telomere biology disorders.

Sources: Teacher: Deoxyribonucleic Acid(DNA Replication (1).pptx, eukaryotic polymerase and telomere sections. Lippincott 6e, Chapter 29, telomere synthesis and reverse transcriptase.

Mismatch, nucleotide-excision, and base-excision repair

Mismatch repair removes replication errors that escape proofreading. Nucleotide-excision repair removes a short oligonucleotide containing a bulky, helix-distorting lesion such as a UV-induced pyrimidine dimer. Base-excision repair begins with a lesion-specific glycosylase that removes a damaged base, followed by AP-site incision, gap filling, and ligation.

- Mismatch repair corrects base mismatches and small insertion/deletion loops.
- NER handles bulky, helix-distorting lesions.
- BER handles small, nonbulky base damage such as deamination.
- All three pathways use an intact complementary strand as sequence information.

Application: Mismatch-repair defects cause Lynch syndrome and microsatellite instability. NER defects cause xeroderma pigmentosum with extreme UV sensitivity and early skin cancers.

Sources: Lippincott 6e, Chapter 29, DNA repair section: mismatch, nucleotide-excision, and base-excision repair. The teacher replication deck is titled DNA replication and the textbook supplies the repair mechanisms needed to complete that chapter outcome.

Homologous recombination versus nonhomologous end joining

Homologous recombination repairs a double-strand break using a homologous template, usually the sister chromatid, and is therefore relatively accurate. Nonhomologous end joining directly processes and ligates broken ends without requiring a homologous template, so sequence may be lost or altered. Failure or misuse of either pathway creates chromosome instability and can initiate malignancy.

- HR is template directed and favored in S/G2 when a sister chromatid is available.
- NHEJ is available more broadly but is intrinsically error prone.
- BRCA1 and BRCA2 participate in homologous recombination.
- End-joining errors can create deletions and translocations.

Application: BRCA1/2 loss increases breast, ovarian, prostate, and pancreatic cancer risk. HR-deficient tumors can be selectively vulnerable to PARP inhibition.

Sources: Lippincott 6e, Chapter 29, double-strand break repair by HR and NHEJ. Lippincott Chapter 29 question extraction includes BRCA and clinical consequences of defective end joining.

Past-paper checkpoints and wording traps

Which of the following enzymes is required for end to end joining of DNA?

DNA ligase is the best answer. Replication requires directional synthesis, proofreading, and coordinated fork enzymes.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q21

Which statement about DNA structure and denaturation is true?

GC-rich DNA has greater thermal stability and a higher melting temperature than AT-rich DNA because GC base pairs provide three hydrogen bonds and stronger stacking interactions.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q57

Xeroderma pigmentosum results from a defect in?

Editorial repair: the source offered bacterial UvrABC as the human xeroderma pigmentosum defect. Human XP usually involves NER proteins XPA-XPB; the XP variant involves POLH/translesion synthesis. UvrABC is bacterial, not a human gene.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q72

Which one of the following nucleic acids has a left-handed helix?

Z-DNA is the best answer. Z-DNA is a left-handed helix.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q73

Okazaki fragments are formed during the synthesis of?

DNA is the best answer. Okazaki fragments are short DNA segments synthesized on the lagging strand.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q8

Ciprofloxacin inhibits.....

DNA gyrase is the best answer. Ciprofloxacin inhibits bacterial DNA gyrase/topoisomerase II, disrupting DNA topology and replication.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q26

Which of the following mechanisms will remove uracil and incorporate the correct base?

Base excision repair is the best answer. DNA repair pathways recognize a lesion, remove it, resynthesize DNA, and seal the strand.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q20

According to Chargaff's rule, in a DNA molecule the percentage of?

$T+G=A+C$ is the best answer. Because $A=T$ and $G=C$ in double-stranded DNA, $T+G$ equals $A+C$.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q23

The enzyme responsible for the removal of supercoiling in replicating DNA, ahead of the replication fork, is?

Topoisomerase is the best answer. Topoisomerase relieves torsional stress ahead of the replication fork.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q33

Which of the following reactions is required for proofreading, during DNA replication, by DNA polymerase III?

3' to 5' exonuclease activity is the best answer. Replication requires directional synthesis, proofreading, and coordinated fork enzymes.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q42

Which of the following leads to disruption of nucleosome structure?

Acetylation is the best answer. Histone acetylation weakens histone-DNA attraction and opens chromatin.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q46

Which is an example of RNA-dependent DNA polymerase?

Reverse transcriptase is the best answer. Reverse transcriptase synthesizes DNA from an RNA template.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q47

The proofreading function of DNA polymerase involves all of the following except

Reversal of the polymerization reaction is the best answer. Proofreading detects a mismatch and removes it with 3'5' exonuclease activity; it is not reversal of polymerization.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q40

When double-stranded DNA is heated at neutral pH, which change does not occur?

The covalent N-glycosidic bond between the base and the pentose breaks is the best answer. Heat disrupts hydrogen bonding and base stacking, but does not cleave the covalent base-sugar N-glycosidic bond.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q4

Histones are _____ that are usually associated with _____ . .

Basic proteins - DNA is the best answer. Histones are basic, lysine/arginine-rich proteins that bind negatively charged DNA.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q33

The compound that consists of ribose linked by an N-glycosidic bond to N-9 of adenine is

Adenosine is the best answer. Adenine plus ribose is the nucleoside adenosine; a phosphate would make it a nucleotide.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q28

In base-excision repair, the first enzyme to act is

DNA glycosylase is the best answer. A DNA glycosylase first removes the damaged base, creating an abasic site.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q9

When a DNA molecule is described as replicating bidirectionally, it means that it has two ...

Replication forks is the best answer. Bidirectional replication produces two forks moving away from an origin.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q35

22 RNA, transcription and processing

Source basis: Lippincott, Chapter 30; concept-specific teacher references below.

RNA types and bacterial versus eukaryotic mRNA architecture

mRNA carries coding information, tRNA is the adaptor that matches codons to amino acids, and rRNA provides structural and catalytic functions in ribosomes. Bacterial mRNA is often polycistronic and can be translated while it is transcribed. Eukaryotic mRNA is usually monocistronic and is processed with a 5' cap, splicing, and a 3' poly(A) tail before export to the cytoplasm.

- mRNA is the translated RNA; tRNA decodes; rRNA forms the catalytic ribosome.
- RNA uses ribose and uracil and is usually single stranded.
- Bacterial transcription and translation can be coupled.
- Eukaryotic pre-mRNA is processed in the nucleus before translation.

Sources: Teacher: Transcription 2025.pptx, RNA-structure and RNA-type introduction. Lippincott 6e, Chapter 30, major RNA types and eukaryotic mRNA architecture.

Primer-independent bacterial transcription initiation

RNA polymerase can begin RNA synthesis de novo and therefore does not require a primer. In bacteria, the sigma factor joins the core enzyme to form a holoenzyme that recognizes promoter consensus sequences. The -35 region supports recognition and the AT-rich -10 Pribnow box facilitates local DNA opening near the +1 start site.

- RNA polymerase synthesizes RNA 5'3' without a primer.
- Sigma confers promoter specificity but is not the elongating catalytic core.
- Canonical bacterial promoter elements are near -35 and -10.
- Promoter sequences are not themselves transcribed.

Application: Rifampicin selectively inhibits bacterial RNA polymerase and is central to tuberculosis therapy.

Sources: Teacher: Transcription 2025.pptx, bacterial RNA polymerase and promoter-initiation block. Lippincott 6e, Chapter 30, bacterial promoter recognition and sigma factor.

Transcript direction, coding/template strands, and bacterial termination

RNA polymerase reads the DNA template strand 3'5' and synthesizes RNA 5'3'. The RNA sequence is complementary to the template and matches the coding strand except that uracil replaces thymine. Intrinsic termination uses a GC-rich RNA hairpin followed by uridines, whereas rho-dependent termination uses an ATP-dependent helicase to release the transcript.

- RNA grows 5'3' and the template is read 3'5'.
- Coding-strand sequence corresponds to RNA with T replaced by U.
- Sigma commonly dissociates after promoter clearance.
- Intrinsic termination requires a hairpin plus a U-rich tract.
- Rho binds RNA and catches a paused polymerase.

Sources: Teacher: Transcription 2025.pptx, elongation and termination sections. Lippincott 6e, Chapter 30, transcript directionality and bacterial termination mechanisms.

Division of labor among RNA polymerases I, II, and III

RNA polymerase I produces the 45S precursor that yields 18S, 5.8S, and 28S rRNAs. RNA polymerase II produces pre-mRNA and several regulatory RNAs. RNA polymerase III produces tRNAs and 5S rRNA, so the exceptional 5S product must not be assigned to polymerase I.

- Pol I: 45S pre-rRNA 18S, 5.8S, 28S.
- Pol II: pre-mRNA and selected sn/miRNAs.
- Pol III: tRNA and 5S rRNA.
- Pol I acts in the nucleolus; Pol II and III act in the nucleoplasm.

Application: α -Amanitin strongly inhibits RNA polymerase II and, at higher doses, polymerase III.

Sources: Teacher: Transcription 2025.pptx, eukaryotic polymerase section. Lippincott 6e, Chapter 30, eukaryotic nuclear RNA polymerases.

RNA polymerase II promoters and general transcription factors

RNA polymerase II does not recognize its promoter alone; general transcription factors assemble a preinitiation complex. TFIID, through TBP, recognizes the TATA box when present, and other core promoter elements can substitute in TATA-less genes. TFIIF opens DNA with helicase activity and phosphorylates the polymerase II C-terminal domain to permit promoter escape.

- TATA/Hogness box is a core promoter element near -25 in many genes.
- TBP is part of TFIID.
- General factors position Pol II and create an open complex.
- TFIIF has helicase and kinase activities.

Application: Defects in TFIIF-associated nucleotide-excision repair/transcription proteins can cause xeroderma pigmentosum or trichothiodystrophy.

Sources: Teacher: Transcription 2025.pptx, Pol II general-factor section. Lippincott 6e, Chapter 30, TATA box, TFIID, and TFIIF.

5' capping and 3' polyadenylation of pre-mRNA

Eukaryotic pre-mRNA receives a 7-methylguanosine cap through an unusual 5'-5' triphosphate linkage early during transcription. The 3' end is cleaved downstream of the AAUAAA signal and poly(A) polymerase adds a tail that is not directly encoded by the DNA template. The cap and tail protect the transcript, support nuclear export, and cooperate in efficient translation initiation.

- The cap is 7-methylguanosine linked 5'-5'.
- The cap supports stability, export, and ribosome recruitment.
- AAUAAA directs cleavage/polyadenylation.
- Poly(A) polymerase adds the non-templated tail.
- Histone mRNAs are an important poly(A)-tail exception.

Application: Mutations in polyadenylation signals can reduce normal mRNA production, including in globin disorders.

Sources: Lippincott 6e, Chapter 30, pre-mRNA capping and polyadenylation. Teacher: Transcription 2025.pptx includes eukaryotic transcription and RNA processing in its confirmed scope.

Spliceosome mechanism, splice-site disease, and alternative splicing

The spliceosome uses snRNPs to recognize the 5' GU site, branch-point adenine, polypyrimidine tract, and 3' AG site. Two transesterification reactions form a lariat and join neighboring exons without consuming ATP for the bond-exchange chemistry itself. Alternative splice-site choice allows one gene to produce different protein isoforms, while pathogenic splice mutations can cause exon skipping, intron retention, or activation of cryptic sites.

- Introns are removed and exons are ligated by the spliceosome.
- The branch-point A forms a 2'-5' bond in the lariat.
- Consensus boundaries are usually GU at the 5' end and AG at the 3' end.
- Alternative splicing expands the proteome without adding genes.

- Splice-site mutations can alter the reading frame or introduce premature stops.

Application: β -thalassemia can result from abnormal β -globin pre-mRNA splicing. Antisense oligonucleotides can therapeutically redirect splicing in selected diseases.

Sources: Lippincott 6e, Chapter 30, spliceosome, splice-site mutations, and alternative splicing. Teacher: Transcription 2025.pptx includes eukaryotic RNA processing in the confirmed course block.

Transcription inhibitors as mechanism probes and drugs

Actinomycin D intercalates into DNA and broadly blocks RNA-chain elongation. Rifampicin binds the β subunit of bacterial RNA polymerase and prevents productive initiation. α -Amanitin from Amanita mushrooms most strongly inhibits eukaryotic RNA polymerase II, shutting down mRNA synthesis.

- Actinomycin D targets DNA and inhibits transcription broadly.
- Rifampicin selectively targets bacterial RNA polymerase.
- α -Amanitin is most potent against Pol II.
- Selectivity depends on whether the target is DNA itself or a domain-specific polymerase.

Application: Rifampicin is a key antimycobacterial drug and a strong hepatic enzyme inducer. Amanita poisoning can produce fulminant hepatic failure.

Sources: Teacher: Transcription 2025.pptx, transcription-inhibitor section. Lippincott 6e, Chapter 30, rifampin, dactinomycin, and α -amanitin.

Past-paper checkpoints and wording traps

The initiation site for transcription is recognized by?

Sigma factor is the best answer. Bacterial sigma factor gives RNA polymerase promoter specificity.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q48

Splice sites are present in?

Eukaryotic hnRNA is the best answer. Splice sites are present in eukaryotic hnRNA/pre-mRNA before introns are removed.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q50

mRNA is complementary to the nucleotide sequence of?

Template strand is the best answer. mRNA is complementary to the DNA template strand and matches the coding strand except U replaces T.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q78

The structure of tRNA appears like a?

Clover leaf is the best answer. The classic two-dimensional structure of tRNA is a cloverleaf.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q82

Binding of the prokaryotic RNA polymerase to promoter site is inhibited by?

Rifampicin is the best answer. RNA polymerase and its factors copy a DNA template into RNA. Rifampicin is the intended drug. More precisely, it binds the bacterial RNA-polymerase beta subunit and blocks early RNA-chain extension, rather than simply preventing promoter recognition.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q3

Actinomycin D is an inhibitor of?

Transcription is the best answer. RNA polymerase and its factors copy a DNA template into RNA.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q15

What is the role of sigma factor in bacterial RNA polymerase?

Positioning RNA polymerase correctly on the DNA template is the best answer. Sigma positions bacterial RNA polymerase at the correct promoter.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q22

Which of the following is an incorrect statement about mRNA?

B and C are both incorrect. Replication-dependent histone mRNAs are capped but usually lack poly-A tails. Splicing removes introns and joins exons. Polyadenylation is typical, not universal.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q26

RNA polymerase from *E. coli* (core enzyme alone) has all of the following properties except that it

Recognizes specific start signals in DNA is the best answer. The bacterial core RNA polymerase can elongate RNA but needs sigma factor to recognize specific promoters.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q50

Which of the followings inhibit the DNA-dependent RNA polymerase in prokaryotes?

Rifampicin is the best answer. Rifampicin inhibits bacterial DNA-dependent RNA polymerase.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q42

23 Translation

Source basis: Lippincott, Chapter 31; concept-specific teacher references below.

Inhibitors: target and process

Agent	Target / action	Process
Rifampicin	Bacterial RNA-polymerase β subunit; early RNA extension	Transcription, not translation
Actinomycin D	DNA intercalation; blocks RNA synthesis	Transcription
Tetracyclines	Prevent aminoacyl-tRNA access to bacterial A site	Translation, 30S
Aminoglycosides	Initiation disruption / misreading	Translation, 30S
Chloramphenicol	Peptidyl-transferase inhibition	Translation, 50S
Puromycin	Aminoacyl-tRNA mimic, premature chain termination	Translation
Ricin	Depurinates 28S rRNA in the 60S subunit	Eukaryotic ribosome inactivation
Diphtheria toxin	ADP-ribosylates eEF2	Eukaryotic translocation
Cycloheximide	Inhibits eukaryotic elongation	Translation, 60S

Genetic code properties, reading frames, and mutation consequences

The genetic code is degenerate because several codons can specify the same amino acid, but it is unambiguous because each codon specifies only one amino acid. An open reading frame extends from an initiation codon through a stop codon in one nucleotide grouping. A nonsense mutation creates a premature stop, while an insertion or deletion not divisible by three shifts the downstream reading frame.

- Degenerate does not mean ambiguous.
- AUG usually initiates translation and also encodes methionine.
- UAA, UAG, and UGA are stop codons.
- Missense changes an amino acid; nonsense creates a stop; frameshift changes downstream codon grouping.
- Silent substitutions can still affect splicing or translation efficiency.

Application: Nonsense and frameshift variants often cause severe loss-of-function genetic disease.

Sources: Teacher: Translation 2025.pptx, slides 6 and 9-10. Lippincott 6e, Chapter 31, genetic code and mutation effects.

Aminoacyl-tRNA synthetases, proofreading, and wobble

An aminoacyl-tRNA synthetase uses ATP to attach a specific amino acid to the 3' CCA end of its cognate tRNA. Many synthetases proofread by removing an incorrectly activated or attached amino acid before the tRNA reaches the ribosome. Flexible pairing between the third codon base and first anticodon base, called wobble, lets one tRNA decode several synonymous codons.

- Each amino acid has a corresponding synthetase specificity system.
- Charging consumes ATP to AMP, equivalent to two high-energy phosphate bonds.

- The ribosome checks codon-anticodon pairing but cannot verify the attached amino acid.
- Wobble explains why fewer tRNAs than sense codons are required.

Application: Some antimicrobial drugs exploit differences in microbial aminoacyl-tRNA synthetases.

Sources: Teacher: Translation 2025.pptx, slides 17-25. Lippincott 6e, Chapter 31, tRNA charging and wobble.

Ribosomal A/P/E sites, peptidyl transferase, and directionality

The ribosome reads mRNA 5'3' and synthesizes protein from the amino terminus toward the carboxyl terminus. Incoming aminoacyl-tRNA enters the A site, peptidyl-tRNA occupies the P site, and deacylated tRNA exits through the E site. The large-subunit rRNA catalyzes peptide-bond formation, demonstrating that the ribosome is a ribozyme.

- A = aminoacyl entry, P = peptidyl chain, E = exit.
- Protein elongates NC.
- Peptide-bond energy comes from the high-energy aminoacyl-tRNA ester bond.
- Bacterial 23S rRNA is the peptidyl-transferase catalyst.

Application: Many antibiotics selectively bind bacterial ribosomal sites or the peptidyl-transferase center.

Sources: Teacher: Translation 2025.pptx, slides 26, 28, and 41-42. Lippincott 6e, Chapter 31, ribosome structure and directionality.

Prokaryotic Shine-Dalgarno alignment versus eukaryotic cap scanning

In bacteria, the Shine-Dalgarno sequence base-pairs with 16S rRNA to position the start codon in the P site. In eukaryotes, the small subunit is recruited near the 5' cap and scans to an AUG in a favorable Kozak context. The initiator tRNA is special because it enters the P site directly; bacteria use formylmethionine whereas cytosolic eukaryotic translation begins with methionine.

- Shine-Dalgarno is bacterial; Kozak/cap scanning is eukaryotic.
- Initiator tRNA starts in the P site, not the A site.
- Bacterial initiation uses fMet-tRNA.
- Initiation factors and GTP assemble the complete ribosome only after correct start recognition.

Sources: Teacher: Translation 2025.pptx, slides 27, 29-30, and 37-38. Lippincott 6e, Chapter 31, prokaryotic and eukaryotic initiation.

Elongation-factor cycle and translocation

During elongation, EF-Tu-GTP delivers an aminoacyl-tRNA to the bacterial A site and GTP hydrolysis helps enforce correct codon recognition. Peptidyl transfer moves the growing chain from the P-site tRNA to the A-site amino acid. EF-G-GTP then drives translocation by one codon, shifting peptidyl-tRNA to P and deacylated tRNA toward E.

- EF-Tu delivers charged tRNA; EF-Ts reloads EF-Tu with GTP.
- Peptidyl transfer precedes translocation.
- EF-G is the bacterial translocase.
- Eukaryotic homologs include eEF1A and eEF2.
- At least two GTP molecules are hydrolyzed per elongation cycle.

Application: Diphtheria toxin and Pseudomonas exotoxin A inactivate eEF2 and stop translocation.

Sources: Teacher: Translation 2025.pptx, slides 39-43. Lippincott 6e, Chapter 31, elongation cycle and energy use.

Stop-codon termination, ribosome recycling, and polysomes

No tRNA recognizes a stop codon; release factors instead enter the A site and stimulate hydrolysis of the completed peptide from the P-site tRNA. GTP-dependent factors then promote release-factor departure and ribosome recycling. Multiple ribosomes can translate one mRNA simultaneously as a polysome, increasing protein output without increasing mRNA number.

- Stop codons are recognized by release factors, not suppressor tRNAs under normal conditions.
- Termination hydrolyzes the peptidyl-tRNA ester bond.
- Ribosomal subunits, mRNA, and tRNA are recycled.
- A polysome is many ribosomes translating one mRNA.

Application: Premature termination codons can trigger nonsense-mediated mRNA decay.

Sources: Lippincott 6e, Chapter 31, termination, recycling, and polysomes. Teacher: Translation 2025.pptx covers the complete translation cycle.

Signal sequences direct proteins to ER, nucleus, and mitochondria

Secreted and many membrane proteins carry an N-terminal signal peptide recognized by signal-recognition particle, which pauses translation and directs the ribosome to the rough ER. Nuclear proteins use nuclear-localization signals and are generally imported after synthesis. Many mitochondrial matrix proteins carry an N-terminal amphipathic targeting sequence and are imported posttranslationally.

- ER targeting is usually cotranslational through SRP and its receptor.
- Free and ER-bound ribosomes are structurally identical; the signal sequence determines destination.
- Nuclear localization signals are not usually cleaved.
- Mitochondrial targeting sequences commonly form amphipathic helices.

Application: Defective peroxisomal targeting causes Zellweger-spectrum disorders. Mistargeting or failed trafficking can cause protein accumulation and organelle dysfunction.

Sources: Lippincott 6e, Chapter 31, protein targeting to ER, nucleus, and mitochondria. Teacher translation deck establishes translation machinery; the book supplies the medically essential destination step.

Posttranslational modification, folding, and ubiquitin-proteasome quality control

Proteins can be cleaved, phosphorylated, glycosylated, hydroxylated, carboxylated, or lipidated after or during translation. N-linked glycosylation begins on asparagine in the ER, whereas O-linked sugars are added mainly to serine or threonine in the Golgi. Persistently misfolded cytosolic proteins are polyubiquitinated and degraded by the proteasome.

- Modification can activate a precursor, target a protein, or change stability.
- N-linked glycosylation is on Asn; O-linked is commonly on Ser/Thr.
- Collagen proline/lysine hydroxylation requires vitamin C.
- Vitamin K supports γ -carboxylation of glutamate in clotting proteins.
- Polyubiquitin commonly marks proteins for proteasomal degradation.

Application: $\Delta F508$ CFTR is often degraded by quality control before reaching the plasma membrane. Vitamin C or K deficiency disrupts specific posttranslational modifications.

Sources: Lippincott 6e, Chapter 31, posttranslational modifications and protein degradation. Lippincott Chapter 31 clinical examples include collagen, clotting proteins, and CFTR quality control.

Antibiotics and toxins mapped to translation steps

Tetracyclines block aminoacyl-tRNA entry into the bacterial A site, whereas macrolides inhibit translocation through the 50S subunit. Puromycin mimics aminoacyl-tRNA and causes premature chain release in both domains. Diphtheria toxin ADP-ribosylates eEF2, and ricin depurinates 28S rRNA, so both arrest eukaryotic protein synthesis by distinct mechanisms.

- Tetracycline: blocks A-site entry on bacterial ribosomes.
- Macrolides: 50S binding and impaired translocation.
- Chloramphenicol: inhibits bacterial/mitochondrial peptidyl transferase.
- Puromycin: aminoacyl-tRNA analog causing premature termination.
- Diphtheria toxin targets eEF2; ricin targets 28S rRNA.

Application: Mitochondrial ribosomes partly resemble bacterial ribosomes, explaining selected antimicrobial toxicities.

Sources: Teacher: Translation 2025.pptx, slides 47-50. Lippincott 6e, Chapter 31, inhibitors of protein synthesis.

Past-paper checkpoints and wording traps

Tetracycline prevents synthesis of polypeptide by:

Preventing binding of aminoacyl tRNA is the best answer. Tetracycline binds the bacterial 30S subunit and blocks aminoacyl-tRNA entry into the A site.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q7

Which of the following inactivates the 60S subunit of eukaryotic ribosomes?

Ricin is the best answer. Ricin depurinates 28S rRNA in the eukaryotic 60S subunit and inactivates it. Ricin irreversibly depurinates 28S rRNA in the 60S subunit. Cycloheximide also inhibits eukaryotic elongation at the large subunit; the intended distinction is ricin-mediated ribosome damage.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q4

Which of the following mediates carrying genetic information from DNA to protein synthesis?

m-RNA is the best answer. mRNA carries coding information from DNA to ribosomes.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q5

What is the main function of t-RNA?

Identifies amino acids and transports them to ribosomes is the best answer. Ribosomes read mRNA through tRNA-mediated amino-acid delivery.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q19

In bacteria the elongation stage of protein synthesis does not involve

IF-2 is the best answer. IF-2 is an initiation factor; elongation uses EF-Tu, GTP, and peptidyl-transferase activity.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q16

Regarding protein synthesis in eukaryotes, all proteins are initially synthesized with?

Methionine at their N-terminus is the best answer. Eukaryotic translation initiates with methionine at the N-terminus.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q3

Which one of the following antibiotics does not function by interfering with the translational process?

Penicillin is the best answer. Penicillin inhibits bacterial cell-wall cross-linking, not translation.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q37

24 Gene regulation

Source basis: Lippincott, Chapter 32; concept-specific teacher references below.

Constitutive versus regulated genes, cis elements, trans factors, and operons

Constitutive genes are expressed relatively continuously for basal functions, whereas regulated genes change expression with conditions, cell type, or signals. A cis-acting element is a DNA sequence that affects linked genes on the same DNA molecule; a trans-acting factor is a diffusible product, usually a protein, that binds a target sequence. A bacterial operon places multiple structural genes under shared regulatory DNA so they can be transcribed together as a polycistronic mRNA.

- Cis = local DNA sequence; trans = diffusible regulator.
- Promoters, operators, enhancers, and response elements are cis elements.
- Repressors, activators, and many transcription factors act in trans.
- Operons coordinate related bacterial genes efficiently.

Application: Pathogenic variants can affect either a coding trans factor or a noncoding cis-regulatory element.

Sources: Teacher: Regulation of Gene Expression 2025.pptx, constitutive/controllable genes, cis/trans factors, and operons. Lippincott 6e, Chapter 32, universal transcriptional-regulation logic.

Lac operon integrates lactose availability with glucose status

Without lactose, active lac repressor binds the operator and blocks transcription. Lactose-derived allolactose binds the repressor and lowers its operator affinity, providing negative-control release. When glucose is low, cAMP rises and cAMP-CAP activates RNA-polymerase recruitment, so maximal expression requires lactose present and glucose absent.

- Allolactose is the inducer that inactivates lac repressor.
- Low glucose high cAMP active CAP.
- Lactose present removes negative control; low glucose provides positive control.
- Maximal lacZYA transcription: lactose present, glucose absent.
- Glucose-mediated suppression is catabolite repression.

Sources: Teacher: Regulation of Gene Expression 2025.pptx, lac-operon block. Lippincott 6e, Chapter 32, lac negative and positive regulation.

Trp operon repression and attenuation sense amino-acid supply

The trp operon is normally on when tryptophan is scarce and is repressed when tryptophan acts as a corepressor. Attenuation adds a second layer by coupling translation of a leader peptide to formation of alternative mRNA hairpins. Abundant charged tRNA-Trp allows the ribosome to advance and favors a terminator, whereas scarcity stalls the ribosome and favors an antiterminator so biosynthetic genes are transcribed.

- Lac is inducible and catabolic; trp is repressible and anabolic.
- Tryptophan activates its repressor as a corepressor.
- Attenuation depends on coupled transcription and translation.
- High Trp favors termination; low Trp permits transcription.

Sources: Teacher: Regulation of Gene Expression 2025.pptx, trp-operon section. Lippincott 6e, Chapter 32, trp attenuation mechanism.

Enhancers, specific transcription factors, and combinatorial control

Specific transcription factors combine a DNA-binding domain with an activation or repression domain. Enhancers can act upstream, downstream, or at long distance and communicate with promoters through DNA looping and coactivator complexes. Different combinations of factors generate cell-specific expression, and genes on different chromosomes can be coordinated if they share response elements recognized by the same trans-acting factor.

- Enhancers are cis DNA elements; activators are trans proteins.
- Enhancer action is often distance and orientation independent.
- Combinatorial control lets a limited factor set specify many expression patterns.
- Shared consensus elements coordinate unlinked genes.

Application: Enhancer mutations and enhancer hijacking can activate oncogenes without altering their coding sequence.

Sources: Teacher: Regulation of Gene Expression 2025.pptx, eukaryotic regulation section. Lippincott 6e, Chapter 32, specific transcription factors and coordinate regulation.

Nuclear receptors and cAMP-CREB connect hormones to transcription

Lipid-soluble hormones can enter cells, bind intracellular nuclear receptors, and direct receptor binding to hormone-response elements such as GREs. Peptide hormones instead signal through cell-surface receptors; glucagon raises cAMP, activates PKA, and phosphorylates CREB. Activated CREB binds cAMP-response elements and recruits transcriptional coactivators to change target-gene expression.

- Nuclear receptors are ligand-activated transcription factors.
- HRE is cis acting; hormone-receptor complex is trans acting.
- Glucagon GPCR cAMP PKA CREB CRE.
- Coactivator versus corepressor recruitment determines transcriptional direction.

Application: Glucocorticoid and thyroid-hormone therapies alter transcription through nuclear receptors. Defects in receptor or response-element signaling can cause hormone resistance.

Sources: Lippincott 6e, Chapter 32, hormone-response elements, nuclear receptors, and CREB. Teacher regulation deck confirms multi-level eukaryotic gene regulation as exam scope.

Histone modifications, chromatin remodeling, and DNA methylation

Gene expression can change heritably without changing DNA sequence through epigenetic mechanisms. Histone acetylation and ATP-dependent chromatin remodeling generally increase access to regulatory DNA, whereas promoter CpG methylation is usually associated with reduced transcription. The pattern is maintained through cell division yet remains more reversible than a DNA mutation.

- Epigenetic change alters expression without changing base sequence.
- Histone acetylation commonly opens chromatin.
- DNA methyltransferases use S-adenosylmethionine as methyl donor.
- Promoter CpG-island methylation usually represses transcription.
- Chromatin remodeling repositions or ejects nucleosomes.

Application: Tumor-suppressor promoters can be silenced by aberrant hypermethylation. DNA-methyltransferase and histone-deacetylase inhibitors are used in selected cancers.

Sources: Teacher: Regulation of Gene Expression 2025.pptx, DNA-methylation and eukaryotic-regulation content. Lippincott 6e, Chapter 32, DNA-level and chromatin-level regulation.

Alternative processing, RNA editing, and mRNA stability

Cells regulate gene output after transcription by selecting splice and polyadenylation sites, editing RNA sequence, controlling nuclear export, and changing mRNA half-life. ApoB editing converts a cytidine to uridine and creates a stop codon, producing shorter ApoB-48 in intestine versus full-length ApoB-100 in liver. Stability elements and RNA-binding proteins determine how long a transcript remains available for translation.

- One pre-mRNA can yield different isoforms through alternative processing.
- RNA editing changes transcript information without changing genomic DNA.
- ApoB-48 is intestinal; ApoB-100 is hepatic.
- Longer mRNA half-life generally permits more protein synthesis.
- Iron-response elements provide a classic stability/translation control system.

Application: ApoB-100 is required for LDL-receptor recognition, whereas ApoB-48 packages dietary lipid in chylomicrons.

Sources: Teacher: Regulation of Gene Expression 2025.pptx, posttranscriptional regulation and ApoB editing. Lippincott 6e, Chapter 32, mRNA processing and stability.

RNA interference and regulation of translation initiation

MicroRNAs are processed into short guide strands loaded into RISC, where base pairing directs translational repression or mRNA cleavage depending on complementarity. Small interfering RNAs use the same core machinery and usually pair extensively with a target for cleavage. Cells also reduce global translation during stress by phosphorylating eIF2, which limits regeneration of the initiation ternary complex.

- Dicer helps generate short RNA duplexes; Argonaute is the RISC effector.
- Partial pairing usually represses translation; extensive pairing favors cleavage.
- siRNA can be experimentally or therapeutically supplied.
- Phosphorylated eIF2 inhibits translation initiation during stress.

Application: Approved siRNA drugs silence pathogenic hepatic transcripts. The integrated stress response uses eIF2 phosphorylation to rebalance protein synthesis.

Sources: Lippincott 6e, Chapter 32, RNA interference and eIF2-mediated translational control. Teacher regulation deck confirms posttranscriptional and translational regulation in the course scope.

Gene rearrangement, amplification, and transposable elements

V(D)J recombination rearranges immunoglobulin gene segments to create enormous antibody diversity from a limited inherited segment set. Gene amplification increases copy number and can raise protein abundance, as in DHFR amplification during methotrexate resistance. Transposons and retrotransposons move through DNA- or RNA-based mechanisms and can disrupt genes or spread antimicrobial-resistance determinants.

- Somatic V(D)J recombination permanently rearranges lymphocyte DNA.
- Copy-number increase can drive overexpression and drug resistance.
- Transposons use transposase; retrotransposons use an RNA intermediate.
- Mobile elements can cause insertional mutation or move resistance genes.

Application: Defective V(D)J machinery causes severe immunodeficiency. Oncogene amplification and mobile resistance cassettes are clinically important forms of genome change.

Sources: Teacher: Regulation of Gene Expression 2025.pptx, immunoglobulin rearrangement in the eukaryotic-regulation section. Lippincott 6e, Chapter 32, gene amplification and transposons.

Past-paper checkpoints and wording traps

The gene of lac operon which has constitutive expression is.....

i is the best answer. lacI is expressed constitutively to supply the repressor protein.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q39

Which of these operons is anabolic?

The trp operon encodes enzymes for tryptophan biosynthesis, so it controls an anabolic pathway and is repressible by its end product. Both tryptophan and phenylalanine biosynthetic operons are anabolic. The original intended trp answer is not uniquely correct when phe is also offered.

Source item: Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q44

Where does a repressor bind to an operon?

Operator is the best answer. Operators, repressors, activators, and feedback control gene expression.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q9

Lactose utilization by E. coli requires which lac gene products?

LacY permease imports lactose and LacZ beta-galactosidase hydrolyzes it. Both A and B are valid for the unrestricted original question.

Source item: Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q12

The operator region normally can be bound by

Repressor is the best answer. A repressor binds the operator and alters transcription of the adjacent structural genes.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q31

The tryptophan operon of E. coli is repressed by tryptophan added to the growth medium. The tryptophan repressor probably

Binds to the trp operator in the presence of tryptophan is the best answer. Tryptophan acts as a corepressor, enabling the repressor to bind the trp operator.

Source item: Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q29

25 Biotechnology

Source basis: Lippincott, Chapter 33; concept-specific teacher references below.

Restriction enzymes, sticky ends, and plasmid cloning vectors

Type II restriction endonucleases recognize short, often palindromic DNA sequences and cut reproducibly. Staggered cuts can create complementary sticky ends that align insert and vector before DNA ligase seals the backbone. A basic plasmid vector needs an origin of replication, a selectable marker, and a cloning site for the inserted DNA.

- Restriction enzymes cut sequence-specific sites.
- Compatible sticky ends improve directional assembly efficiency.
- DNA ligase covalently seals vector and insert.
- Origin, selectable marker, and cloning site are core vector features.

Application: Recombinant plasmids enable production of therapeutic proteins such as human insulin.

Sources: Lippincott Illustrated Reviews: Biochemistry, 6e, Chapter 33, PDF pp. 838-877. Tracked bank source labels Chapter 33 as confirmed August 25 Cell & Molecules scope.

cDNA and complementary nucleic-acid probes

Reverse transcriptase copies mature mRNA into complementary DNA, so cDNA represents expressed, already-spliced sequence and lacks genomic introns. That property allows a human coding sequence to be expressed in bacteria, which cannot remove most eukaryotic introns. A labeled probe detects a target by complementary base pairing under controlled stringency.

- cDNA reflects the transcripts present in the sampled cell.
- Mature-mRNA-derived cDNA lacks introns.
- Reverse transcriptase makes the initial DNA copy.
- Probe specificity depends on complementarity, temperature, salt, and mismatch tolerance.

Application: cDNA and probes are used to measure gene expression and detect infectious or inherited sequences.

Sources: Lippincott 6e, Chapter 33, cDNA synthesis and nucleic-acid hybridization. Confirmed August 25 Chapter 33 question bank, PDF pp. 838-877.

PCR amplification and reverse-transcription PCR

Each PCR cycle proceeds through strand denaturation, primer annealing, and thermostable-polymerase extension. The two primers bind opposite strands with their 3' ends aimed toward the target, thereby defining the amplified interval. To analyze RNA, reverse transcriptase first creates cDNA, which then serves as the PCR template.

- Cycle order: denature anneal extend.
- Primers determine specificity and amplicon boundaries.
- Thermostable polymerase survives repeated heating.
- RT-PCR begins with reverse transcription of RNA.
- Quantitative PCR follows amplification using fluorescence.

Application: PCR is used for pathogen detection, genotyping, minimal-residual-disease testing, and viral-load assays.

Sources: Lippincott 6e, Chapter 33, PCR and reverse-transcription PCR. Confirmed August 25 Chapter 33 question bank, PDF pp. 838-877.

Southern, Northern, and Western blot targets

Southern blotting detects a specific DNA sequence after fragment separation and transfer to a membrane. Northern blotting detects RNA transcripts and can report transcript size and relative abundance. Western blotting detects a protein using target-specific antibodies after electrophoresis and membrane transfer.

- Southern = DNA with a nucleic-acid probe.
- Northern = RNA with a nucleic-acid probe.
- Western = protein with antibodies.
- All separate a sample, transfer it, and detect a specific target.

Application: Western-type immunoblots can confirm protein identity or pathogen-specific antibodies, though many clinical workflows now use other platforms.

Sources: Lippincott 6e, Chapter 33, blotting methods. Confirmed August 25 Chapter 33 question bank, PDF pp. 838-877.

Restriction-fragment polymorphism and Sanger sequencing

A variant that creates or abolishes a restriction site changes fragment lengths and can be detected as an RFLP after digestion and suitable fragment analysis. Sanger sequencing uses occasional dideoxynucleotide incorporation to terminate extension because ddNTPs lack a 3'-OH. Sorting terminated fragments by length reveals the newly synthesized DNA sequence.

- RFLP detects sequence-dependent fragment-length differences.
- ddNTP incorporation prevents the next phosphodiester bond.
- Sanger products differ by one nucleotide in length.
- Sequence is read from the order of labeled terminated fragments.

Application: Sanger sequencing remains a common confirmatory method for targeted variants.

Sources: Lippincott 6e, Chapter 33, RFLP analysis and chain-termination sequencing. Confirmed August 25 Chapter 33 question bank, PDF pp. 838-877.

Expression microarrays and allele-specific hybridization

A gene-expression microarray hybridizes labeled cDNA from a sample to thousands of fixed probes, allowing relative transcript abundance to be compared across conditions. An allele-specific oligonucleotide can distinguish sequences differing by one nucleotide if hybridization stringency preserves the perfect match but destabilizes the mismatch. Both methods require careful controls because signal measures hybridization, not necessarily protein abundance or function.

- Microarray signal estimates relative mRNA abundance through cDNA.
- Allele-specific probes depend on mismatch-sensitive stringency.
- Hybridization results require normalization and suitable controls.
- Expression change does not prove a causal functional change.

Application: Expression signatures can classify selected tumors and predict biologic behavior.

Sources: Lippincott 6e, Chapter 33, microarray and allele-specific detection methods. Confirmed August 25 Chapter 33 question bank, PDF pp. 838-877.

26 Laboratory principles and tests

Source basis: Teacher foundations / laboratory material; concept-specific teacher references below.

Bench-test discrimination

Test / method	Detects or measures	Interpretation caution
Benedict / Fehling	Reducing sugars	Not specific to glucose; sucrose is negative before hydrolysis.
Barfoed	Monosaccharides reduce reagent relatively rapidly	Heating time matters; prolonged heating blurs the distinction.
Seliwanoff	Ketoses react rapidly in acid	A timed distinction; sucrose can yield fructose after hydrolysis.
Iodine	Starch-like helical polysaccharide complexes	Not a general test for all carbohydrates.
Biuret	Peptide-bond copper complex	Free amino acids do not give the protein reaction.
Ninhydrin	Free amino groups	Most amino acids purple; proline gives a different yellow response.
Beer-Lambert law	$A = \epsilon lc$; $A = -\log_{10}(T)$	Blank correctly; use linear range and matching wavelength/path length.
Chromatography	Differential partition/interaction	$R_f = \text{solute distance} / \text{solvent-front distance}$ in planar chromatography.

General carbohydrate, starch, and reducing-sugar tests

Molisch is a broad carbohydrate test that produces a violet ring at the acid-aqueous interface. Iodine forms a blue-black complex with the helical structure of starch and is useful for polysaccharide recognition. Benedict reagent detects reducing sugars by reduction of Cu^{2+} to Cu_2O , with color progressing from green toward orange or brick red as reducing-sugar concentration increases.

- Molisch: general carbohydrate; violet interfacial ring.
- Iodine: starch/polysaccharide; blue-black complex.
- Benedict: reducing sugar; colored Cu_2O precipitate after heating.
- A positive broad test does not identify the exact carbohydrate.

Application: Copper-reduction tests are historical screening methods; modern glucose assays are more specific and quantitative.

Sources: Teacher: Carbohydrate Qualification Test Esmaeili Jan 2026.pdf, pp. 7-15. Telegram scope audit identifies this 21-page file as confirmed practical biochemistry material.

Use reaction conditions and timing to distinguish carbohydrate classes

Benedict color is semiquantitative because increasing reducing-sugar concentration produces progressively more colored Cu_2O precipitate rather than an exact calibrated concentration. Barfoed uses acidic copper reagent, in which monosaccharides reduce copper rapidly while disaccharides react more slowly. Seliwanoff uses acid and resorcinol; ketoses dehydrate rapidly and form a red product sooner than aldoses.

- Benedict estimates reducing-sugar amount only roughly by color.

- Barfoed separates rapid monosaccharide reduction from slower disaccharide reduction.
- Seliwanoff favors rapid ketose detection.
- Timing matters: prolonged heating can create misleading late positives.

Sources: Teacher: Carbohydrate Qualification Test Esmaeili Jan 2026.pdf, pp. 14-19. Teacher-derived image spotters confirm the expected Benedict, Barfoed, and Seliwanoff appearances.

Ninhydrin, Biuret, and xanthoproteic reaction targets

Ninhydrin reacts with free amino groups and gives Ruhemann's purple for most amino acids, while proline characteristically gives yellow. Biuret detects multiple peptide bonds through a violet Cu^{2+} complex in alkaline solution and is therefore a protein/peptide test rather than a free-amino-acid test. The xanthoproteic reaction nitrates aromatic residues, producing yellow that deepens toward orange after alkali.

- Ninhydrin: free amino groups; purple for most amino acids.
- Proline is the yellow ninhydrin exception.
- Biuret: Cu^{2+} complex with multiple peptide bonds in alkali.
- Xanthoproteic: aromatic amino-acid residues; yellow-to-orange product.

Application: Modern clinical protein quantification uses standardized colorimetric or immunochemical assays, but Biuret chemistry remains widely used for total protein.

Sources: Confirmed TUMS Cell & Molecules practical curriculum and data/teacher-materials/exam-scope.md. Tracked curriculum-gap bank: Practical amino acid and protein tests. No separate local teacher deck is recorded in the current Telegram scope audit.

Precision glassware, meniscus reading, and burette delivery

A volumetric pipette is calibrated to deliver one specified volume with high precision, whereas a graduated pipette can deliver variable volumes with lower precision. A typical aqueous meniscus should be read at eye level from its bottom to avoid parallax. A burette delivers controlled variable volumes of titrant through a stopcock and is read from the change between initial and final graduations.

- Volumetric pipette: one precise delivered volume.
- Volumetric flask: one precise contained final volume.
- Read the lower aqueous meniscus at eye level.
- Burette: variable controlled delivery for titration.
- Correct pipette fillers are used; mouth pipetting is never acceptable.

Application: Preanalytical volume errors can bias every downstream calculated concentration.

Sources: Teacher: Laboratory Equipment Esmaeili Jan 2026.pdf, pp. 18-27. Telegram scope audit confirms the 46-page laboratory-equipment deck.

Recognize core equipment and match it to safety or separation function

A fume hood captures hazardous vapors and exhausts them away from the worker; it is not a sterile cabinet. An analytical balance uses a draft shield to support high-precision mass measurement. A centrifuge accelerates sedimentation so components separate according to a combination of size, shape, and density under the applied relative centrifugal force.

- Fume hood protects from chemical vapors; it does not guarantee product sterility.
- Analytical balance doors reduce air-current error.
- Centrifugation separates by sedimentation behavior, not pH alone.
- Rotors must be balanced by mass and tube position before spinning.
- Use relative centrifugal force when protocols require cross-instrument reproducibility.

Application: Improper centrifugation can hemolyze samples or leave cells in plasma/serum and distort results.

Sources: Teacher: Laboratory Equipment Esmaeili Jan 2026.pdf, pp. 36 and 43-44. Teacher-derived image spotters identify analytical balance, fume hood, and centrifuges.

Acid-base titration endpoint and normality calculation

In the teacher experiment, standardized 0.1 N NaOH is delivered from the burette to determine an unknown acid concentration. Phenolphthalein marks the endpoint as the first faint pink color persisting for roughly 30 seconds; a strong pink indicates overshooting. At the stated equivalence relationship, unknown normality follows from $N_1V_1 = N_2V_2$ using consistent volume units.

- Titrant is known; analyte is unknown.
- Teacher standard: 0.1 N NaOH.
- Endpoint is persistent pale pink, not deep pink.
- Normality counts reactive equivalents per liter.
- Record initial and final burette readings and use their difference.

Application: Titrimetric principles underlie several classical laboratory standardizations and acid-base measurements.

Sources: Teacher: Titration Esmaeili Jan 2026.pdf, pp. 6-9. Telegram scope audit confirms this 10-page practical titration file.

Initial velocity, reagent blanks, and enzyme units

Enzyme activity is measured in the early linear phase so substrate is not yet substantially depleted and product accumulation has not driven the reverse reaction. A reagent blank contains the nonenzymatic assay components but lacks active enzyme or otherwise prevents reaction, allowing background signal to be subtracted. One conventional enzyme unit is the amount catalyzing conversion of one micromole of substrate per minute under defined conditions.

- Use initial velocity in the linear range.
- A blank measures non-analyte/reagent background.
- Activity depends on temperature, pH, substrate concentration, and assay time.
- An enzyme unit is a rate under specified conditions, not a mass.
- Boiling usually destroys activity by denaturing protein structure.

Application: Serum ALT, AST, ALP, CK, and lipase are commonly reported as catalytic activity rather than enzyme concentration.

Sources: Confirmed TUMS Cell & Molecules practical curriculum and data/teacher-materials/exam-scope.md. Tracked curriculum-gap bank: Practical enzyme assays. No separate local teacher deck is recorded in the current Telegram scope audit.

Beer-Lambert law, blanking, and wavelength selection

Within its linear range, Beer-Lambert law states $A = \epsilon bc$, so absorbance is proportional to molar absorptivity, path length, and concentration. A reagent blank removes absorbance caused by solvent, reagents, and cuvette so the remaining signal is attributed to analyte or reaction product. Measuring near the wavelength of maximum absorbance improves sensitivity and reduces the effect of small wavelength-setting errors.

- $A = \epsilon bc$ within the validated linear range.
- Absorbance is $-\log_{10}$ of fractional transmittance.
- Blank composition should match the sample except for the measured analyte/reaction.
- Use $\lambda_{\max}$ for sensitivity.
- Dilute and apply the dilution factor when absorbance exceeds the calibration range.

Application: Many glucose, bilirubin, protein, and enzyme assays convert concentration or reaction rate into absorbance.

Sources: Confirmed TUMS Cell & Molecules practical curriculum and data/teacher-materials/exam-scope.md. Tracked curriculum-gap bank: Spectrophotometry. No separate local teacher deck is recorded in the current Telegram scope audit.

DNA extraction by lysis, nuclease inhibition, and alcohol precipitation

Detergent disrupts lipid membranes and helps release DNA from cells and nuclei. EDTA chelates divalent cations required by many nucleases, while protease removes protein contaminants including histones and nucleases. Salt shields phosphate charges and cold ethanol lowers DNA solubility, allowing DNA to aggregate and precipitate from the aqueous extract.

- Detergent lyses membranes.
- EDTA chelates Mg^{2+}/Ca^{2+} and suppresses nucleases.
- Protease removes DNA-bound and contaminating proteins.
- Salt plus cold alcohol precipitates DNA.
- Avoid vigorous shearing when intact high-molecular-weight DNA is needed.

Application: Poor extraction quality can cause false-negative PCR or failed sequencing.

Sources: Confirmed TUMS Cell & Molecules practical curriculum and data/teacher-materials/exam-scope.md. Tracked curriculum-gap bank: DNA extraction. No separate local teacher deck is recorded in the current Telegram scope audit.

Chromatographic separation by mobility, charge, size, or affinity

In paper or thin-layer chromatography, R_f is the distance traveled by solute divided by the solvent-front distance under the same run conditions. Ion-exchange chromatography separates by net charge through reversible binding to oppositely charged groups. Affinity chromatography uses a specific immobilized ligand to retain a target, while size-exclusion chromatography makes larger molecules elute first because they enter fewer bead pores.

- R_f is unitless and lies between zero and one in a valid planar run.
- Ion exchange separates principally by net charge.
- Affinity uses specific target-ligand recognition.
- Size exclusion: large molecules elute before small molecules.
- Changing pH or ionic strength can alter binding and elution.

Application: Chromatographic methods separate hemoglobin variants, drugs, metabolites, and proteins in clinical laboratories.

Sources: Confirmed TUMS Cell & Molecules practical curriculum and data/teacher-materials/exam-scope.md. Tracked curriculum-gap bank: Chromatography. No separate local teacher deck is recorded in the current Telegram scope audit.

Atomic emission, calibration, and clinical ions in flame photometry

A flame excites atoms, and light emitted as electrons return to lower energy levels is measured at element-specific wavelengths. Sodium, potassium, and lithium are common analytes because their atomic emissions are strong and characteristic. An unknown concentration is obtained from standards and a calibration curve; out-of-range samples must be diluted into the validated range and corrected by the dilution factor.

- Flame photometry measures emitted light, not absorbance.
- Emission wavelength identifies the element; intensity relates to concentration.
- Na^+ and K^+ are classic analytes.
- Standards and blanks establish a valid calibration.
- Never extrapolate far beyond the highest standard.

Application: Flame emission was widely used for serum sodium and potassium and remains conceptually important in electrolyte analysis.

Sources: Confirmed TUMS Cell & Molecules practical curriculum and data/teacher-materials/exam-scope.md. Tracked curriculum-gap bank: Flame photometry. No separate local teacher deck is recorded in the current Telegram scope audit.

Osmolarity, tonicity, RBC behavior, and dialysis

Hypotonic extracellular fluid causes red cells to gain water, swell, and potentially hemolyze, whereas hypertonic fluid causes water loss and crenation. Tonicity depends on effective nonpenetrating solutes rather than total osmolarity alone, so an isosmotic solution of a readily permeant solute such as urea can still be hypotonic. Across a dialysis membrane, water initially moves toward the compartment with the higher concentration of retained solute.

- Hypotonic cell swelling/hemolysis.
- Hypertonic cell shrinkage/crenation.
- Isosmotic does not always mean isotonic.
- Penetrating solutes contribute little to sustained tonicity after equilibration.
- Water moves toward higher effective osmolarity across a water-permeable membrane.

Application: Incorrect IV-fluid tonicity can cause dangerous cellular swelling or dehydration, especially in the brain.

Sources: Confirmed TUMS Cell & Molecules practical curriculum and data/teacher-materials/exam-scope.md. Tracked curriculum-gap bank: Osmosis practical. No separate local teacher deck is recorded in the current Telegram scope audit.

Sources and scope audit

1. Latest original course-scope reconciliation, 20 August 2026; original syllabus and teacher-material archive.
2. Ferrier, D. R. Lippincott Illustrated Reviews: Biochemistry, 6th edition (2014): confirmed chapters listed above.
3. Teacher foundations: INTRODUCTION.ppt; Water and Buffer 1404; Marks Essentials water/buffers reference.
4. Teacher biomolecules: Amino acids and Proteins Parts 1-2; Lipid structure (Esmaeili); Vitamin 2024.
5. Teacher enzymes and genetics: Enzyme Kinetics and Regulation; DNA Structure; DNA Replication; Transcription 2025; Translation 2025; Regulation of Gene Expression 2025.
6. Teacher laboratories: Laboratory Equipment, Titration and Carbohydrate Qualification Test (Esmaeili, January 2026), plus the original laboratory scope.
7. Four source-paper biochemistry sections: April 2021 Cell & Molecules report, February 2021, September 2021, Biochemistry 1 Finals 2022. Filename dates are retained; a report date is not proof of the exam sitting date.
8. Source caveats: human DNA repair - <https://www.ncbi.nlm.nih.gov/books/NBK1397/> ; enzyme kinetics - <https://www.ncbi.nlm.nih.gov/books/NBK92007/> ; hepatic HDL/VLDL - <https://www.ncbi.nlm.nih.gov/books/NBK351/> .

Coverage limits

This review covers the confirmed original biochemistry scaffold and the imported four-paper biochemistry sections. It does not certify a new retake announcement, predict future questions, include other-subject questions, or claim that the archive contains every historical sitting. Original textbook chapters remain the detailed reference.